Decentralized Crypto World Bank

A Blockchain-Based Lending Platform
with AI-Enhanced Security

by

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A pre-thesis 1 report submitted to the Department of Computer Science and Engineering
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Declaration

It is hereby declared that

1. The report submitted is our own original work while completing degree at BRAC University.
2. The report does not contain material previously published or written by a third party, except where this is appropriately cited through full and accurate referencing.
3. The report does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
4. We have acknowledged all main sources of help.

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Ethics Statement

This project operates exclusively on public blockchain test networks using test tokens with no real monetary value. No real financial transactions, personal banking data, or user financial records are involved at any stage of development or evaluation. All transaction data used for Artificial Intelligence and Machine Learning (AI/ML) model training and evaluation is either synthetically generated or sourced from publicly available anonymized datasets. The platform prototype does not process Know Your Customer (KYC) or Anti-Money Laundering (AML) data in its current scope, and all wallet addresses used during testing are disposable testnet addresses. We acknowledge the ethical considerations inherent in AI-assisted financial decision-making and have incorporated explainability mechanisms (SHapley Additive exPlanations, or SHAP) to ensure that automated risk assessments remain transparent and auditable by human reviewers.

Abstract

Global development finance relies on multilayered institutional structures to distribute capital across borders and communities, yet these systems often struggle with fragmented transparency, procedural friction, and uneven access to credit. Although decentralized finance has demonstrated that programmable infrastructures can automate lending and enhance auditability, existing models largely employ flat architectures that do not reflect institutional hierarchies or structured governance. As a result, there remains limited exploration of how tiered financial systems might be represented within decentralized environments while preserving oversight and adaptability. Here we present *Crypto World Bank*, a prototype framework that models a multi-tier institutional lending architecture on a programmable blockchain infrastructure. We show that hierarchical capital flows, role-based governance, and data-informed risk analytics can be coordinated within a unified smart contract environment, enabling transparent state visibility alongside adaptive decision support. This hybrid design illustrates how institutional finance and decentralized systems may converge, suggesting a pathway toward more accountable, flexible, and inclusive digital credit ecosystems.

Keywords: Blockchain, Decentralized Finance, Institutional Architecture, Financial Inclusion, Smart Contracts, AI-Augmented Governance

Dedication

Dedicated to our families for their unwavering support throughout our academic journey.

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Introduction

The combination of blockchain technology, decentralized finance, and artificial intelligence creates new opportunities to address historic inefficiencies in the global development finance. The Crypto World Bank is a project, which takes into consideration how an institutional framework of a decentralized lending system can make more transparent, reduce settlements friction, and reach under-served credit markets. The complete source code, smart contracts, and supporting material for this project are available in our GitHub repository.

Finance across borders is still limited by sluggish correspondent-banking settlement, piecemeal ledger visibility and costly transfer. International remittances have approximated about \$860 billion annually by 2024, but \$48–56 billion is lost in the transfer fees [26]. At the same time, the number of unbanked adults in the world remains approximately 1.4 billion [14], and the gap in financing micro, small and medium enterprises is also more than \$4.5 trillion [20]. At the same time, decentralized finance has shown the logic of lending to be transparently implemented on blockchain infrastructure, although existing leading protocols continue to utilize flat and pool-based designs that fail to reflect the institutional hierarchy and directional capital flows of development finance.

We put blockchain here as a coordination technology, which provides visibility of shared states, audit trails that are difficult to alter, and enforcement of programs among different institutions. Governance design, security controls, and regulatory considerations are also taken into account on the platform according to a staged implementation roadmap incorporating academic validation and regulatory sandbox pilots. Another recent example of a hybrid institutional structure in finance is the Crypto World Bank, which suggests four-layer institutional hierarchy coupled with an open ledger infrastructure and a lightweight AI-assisted monitoring layer, to enhance efficiency in operations and access to credit in emerging markets.

1.1 Background

Capital is provided to local lenders through multilateral overlays of institutional arrangements where multilateral development institutions (e.g., World Bank, International Monetary Fund), national financial intermediaries, and local lenders collaborate to play a role in financing multilateral development projects to local borrowers end borrowers. Although this model allows risk sharing and penetrating local markets, it is linked to great operational issues. In cross-border correspondent banking, financial institutions are obligated to have pre-funded nostro and vostro accounts at correspondent banks in every currency corridor, which maroon huge amounts of capital in idle balances unable to be invested or lent out [24]. Crossborder transactions are regularly settled in two to five business days with an average cost of about $42 per transaction by using the correspondent banking network [25]. The remittance market, which is valued at about $860 billion annually, loses between $48–56 billion per annum to transfer charges, with the average cost to transfer in the world being 6.49% as of 2024, more than the United Nations Sustainable Development Goal limit of 3% [26].

At the same time, it is estimated that there are 1.4 billion unbanked adults in the world as defined by the World Bank Global Findex [14], and most of them are found in the developing economies where documentation, geographic location, and minimum balance requirements leave out high populations of people in the formal financial sector. The International Finance Corporation estimates that there is a financing gap in the world of micro, small and medium enterprises at $4.5 trillion per year [20], which is unmet credit demand that cannot be effectively met by the existing institutions through the conventional intermediation chains.

The DeFi industry has already shown that lending, collateral management, and interest calculation can be automated on a smart contract platform and that all transactions are transparent, on-chain. The largest DeFi lending protocol, Aave v3, has amassed more than $26 billion in total value locked in ten blockchain networks [27], and the DeFi lending market has more than $55 billion in TVL [13]. Nevertheless, these protocols utilize flat, pool based designs where all the lenders feed into one pool and all borrowers withdraw off the same source—irrespective of institutional status, creditworthiness and geographies. There is no current DeFi scheme that reflects the multi-tier institutional structure, inter-tier capitals movement and tiered borrower access provision that is typical of real-world development finance.

1.2 Rationale of the Study

The combination of three forces, namely: (1) the long-standing inefficiencies in the traditional development finance, which restrict the transparency and speed of settlement; (2) the lack of transparency in the application of blockchains to multi-tier institutional lending to DeFi; (3) the possibility of integrating blockchain auditability with lightweight analytics and monitoring support to enhance operational control, drives this study. With an architecture that will bridge these areas, we hope to show a plausible way forward toward transparent, programmable, and institutionally significant lending to underserved populations.

1.3 Problem Statement

The international system of the development finance works based on a stratified system in which funds are directed down by large supranational organizations such as the World Bank and IMF to national institutions, which transmit them to regional and local banks, and they are ultimately delivered to individual borrowers. This intermediary chain brings into being a number of significant inefficiencies:

1. **Lack of transparency.** The financial documents at the various levels are not openly accessible. Individuals lower down the ladder do not understand much of how money is managed, what reserve levels appear like, or what lending decisions appear like at higher levels. The reserve ratios of banks are mostly self-reported, and audit occurs at most quarterly, i.e. there is no means of checking up on financial health in real time.
2. **Sluggish settlements and unproductive capital.** Transmission of money across the borders necessitates going through many middlemen, verification of compliance, and verification steps that are capable of slowing down transactions by two to five business days. Banks must also have pre-funded accounts in partner institutions in every currency in which they trade, communicating large sums of money that would otherwise be engaged in productive employment [24]. An average transaction via the correspondent banking system costs approximately $42, which hits small transactions and participants the most in developing economies [25].
3. **Inconsistent risk evaluation.** Fraud checks and credit evaluations are dependent on each other on human judgment, which develops bias, inconsistency, and limited scalability. Due to the lack of a unified and verifiable credit history, borrowers have to be re-examined introducing extra time and cost at each stage of the chain.
4. **Obstacles to access and institutional trust.** Developing inter-institutional trust through legal agreements, compliance processes and audits are expensive and time-consuming, putting the smaller institutions and borrowers of the developing countries in particular disadvantage. The world at large is unequal, and the situation is set to worsen, according to the World Inequality Report 2026, the global monetary system continues to divert resources out of developing economies to the richer ones, and developing countries experience differentials of returns three percentage points lower than in developed economies [28]—a structural gap that makes accessing international credit markets even more difficult.

At the same time, the DeFi ecosystem has shown that the lending, collateral management, and interest calculation can be completely automated through smart contract platforms with complete transparency. The TVL of Aave v3 is $26.3 billion and active borrows are $17.7 billion [27]; the TVL of Compound v3 is $1.4 billion [29]; and the combined DAI and USDS stablecoin issuance of MakerDAO (renamed Sky Protocol) is approximately $10.5 billion with projected 2026 revenue of $611 million [30]. Despite this scale, these protocols exhibit fundamental limitations in the context of institutional development finance:

- They employ **flat, peer-to-peer**, or **over-collateralized** lending models, where everyone and large institutional borrowers, along with individual retail clients, are dealing with the undifferentiated liquidity pool. No existing protocol models institutional hierarchy, tiered capital allocation or cross-tier lending flows in as similar a manner as the World Bank → National Bank → Local Bank model of development finance.
- They are not connected with **AI-based risk analytics** and **explainable decision support** systems. DeFi lending decisions are determined solely using collateral ratio and utilization curve and no behavioral fraud detection, anomaly detection or explainable credit judgement is present.
- They do not model **role based, tiered governance** of the development finance institutions. It is a form of leadership, in which a majority of the large token holders will wield the power of decision-making, without the institutional role distinction (regulator, wholesale lender, retail lender, borrower) of hierarchical finance.
- The credit-based and emerging-market lending has also been introduced with other new institutional DeFi projects such as Maple Finance ($2.6–3.8 billion TVL, undercollateralized institutional lending) [31] and Goldfinch ($680 million loan originations in 18+ developing countries) [32], however, are single-level projects with no hierarchical capital flows and no interbank lending facilities.

1.4 Objectives

The objectives of this project are:

1. To develop and deploy a four-level lending architecture (World Bank → National Bank → Local Bank → Borrower) on an EVM-capable blockchain that maintains institutional hierarchy and provides shared access to the ledger.
2. To investigate a lightweight off-chain analytics support layer, including fraud detection, anomaly detection, and explainable review support, to supplement human credit governance.

3. To demonstrate transparent and programmable lending processes with configurable borrowing limits, installment payments, and role-based access control implemented through smart contracts.
4. To test the prototype on public testnets (e.g., Polygon, Ethereum Sepolia) and verify hierarchical controls, transparency properties, and operational workflows.
5. To document the design of governance mechanisms, security controls, regulatory considerations, and a controlled rollout pathway toward academic validation and potential regulatory sandbox piloting.

1.5 Research Questions

The research questions used in the study are as follows:

1. **RQ1:** Does a hierarchical blockchain-based lending architecture reflect more faithfully the real-world capital flow mechanism of development finance than present decentralized lending?
2. **RQ2:** Can on-chain transparency, programmable reserve controls and role-based governance system make settlements less opaque and less frictional across institutional tiers?
3. **RQ3:** Does a lightweight off-chain analytics layer offer practical and auditable support to monitor fraud and lending in such a system?
4. **RQ4:** Does the proposed architecture work technically and operationally on current test networks in the public and yet retain their relevance to the real-life institutional and financial-use cases?

1.6 Blockchain Justification

The multi-party coordination and trust problem of incompatible institutions with potential conflicting interests is the root cause of the above problem. We argue that blockchain technology is the solution to this kind of problems than its traditional counterparts due to the following reasons:

1. **Minimizing trust based consensus.** Distributed consensus algorithms make sure that no one can change the ledger state unilaterally and a trusted central operator is not required.
2. **Programmable enforcement.** The rules of lending, such as limits on borrowing, installment payments and workflows are stored in smart contracts as deterministic self-executable programs, reducing the usage of manual processes and bilateral contracts.
3. **Cryptographic auditability.** Each change of state is recorded in an immutable and publicly verifiable record of transactions and will be subject to real-time

auditing by regulators, partners, and borrowers without expensive third-party interactions.

4. **Composable incentive structures.** Chain-based reputation systems, governance tokens, and programmable distributions of fees can bring incentives into harmony across the whole group of network participants.

The convention cloud-based data architecture would impose ledger integrity, access control and audit assurances to a trusted central operator, which would reestablish the identical trust reliance that this project aims to eliminate. Blockchain does away with this failure of trust.

1.7 Proposed Solution

The Crypto World Bank is based on a **four tier on-chain lending model**:

- **Tier 1 — World Bank:** Custodian of the global crypto reserve. Allocates capital to registered National Banks via lending mediated by smart contracts.
- **Tier 2 — National Banks:** Borrow from the World Bank reserve. Lend to Local Banks incorporated in their jurisdiction. Aggregate the risk exposure at the national level.
- **Tier 3 — Local Banks:** Borrowing of National Banks. Loan applications by process originators from individual borrowers. Administer loan lifecycle using particular approvers.
- **Tier 4 — Borrowers:** Loan requests to Local Banks. Repay through configurable installment schedules. Construct on-chain payoff history which determines borrowing in the future.

The platform also incorporates:

- **Risk-sensitive borrowing limits** computed from rolling 6-month and 1-year transaction windows.
- **Automatic installment generation** for loan amounts exceeding a configurable threshold (e.g., 100 ETH equivalent).
- **Off-chain AI/ML security analytics** (planned) for fraud detection (Random Forest), anomaly identification (Isolation Forest), and explainable risk assessment (SHAP).

1.7.1 Cross-Tier Lending System

Capital does not necessarily flow down in the old banking system. Banks at the same level deposit with one another regularly in the interbank lending market (e.g., the federal funds market) and less significant banks that have surplus cash inject the extra cash upwards with their home institutions or central banks. The Crypto World Bank is the extension of its hierarchical architecture to accommodate such **multi-directional lending flows**, developing a more realistic and robust financial ecosystem.

Same-Tier Lending

Banks at the same hierarchical level can lend to one another to manage short-term liquidity:

- **National Bank ↔ National Bank:** A national bank, which has excess reserves, can lend to a second party that is in a liquidity crunch, similar to the traditional interbank lending market. The lending rate is adjusted according to the supply and demand through national banks of the system, like the Secured Overnight Financing Rate (SOFR).
- **Local Bank ↔ Local Bank:** Local banks in same or different national jurisdictions can share liquidity through a peer lending pool. This prevents local liquidity crunches when one bank possesses surplus deposits and the other is experiencing elevated loan demand.

Same-tier lending is done by an on-chain **InterBankLendingPool** contract at every tier level in which surplus banks provide short-term liquidity and those banks that need it are allowed to borrow at utilization floating rates. In the current prototype, the InterBankLendingPool is specified at the design level; the implementation focuses on the primary top-down capital flow (World Bank → National Bank → Local Bank → Borrower), with same-tier and upward lending reserved for subsequent development stages.

Upward Lending

Banks with excess capital of lower tier are then able to lend up the ladder:

- **Local Banks → National Banks:** When local banks accumulate reserves beyond the minimum reserve ratio, they are allowed to deposit excess capital in their parent national bank's liquidity pool with earned deposit yield. This mirrors how commercial banks save the surplus with central banks by means of standing deposit facilities.
- **National Banks → World Bank:** National banks may provide excess to the international deposit, enhancing the total capital base of the system. This provides a back route to well capitalized national banks and raises the reserve available for downward distribution to banks with higher demand.

The downward lending rates are higher than the upward lending rates (the low risk of lending to a higher-level institution), developing a natural interest rate term structure across the hierarchy.

Tiered Borrower Access

Different borrower classes access different tiers based on loan size and organizational type:

Table 1.1: Borrower tier access rules.

Borrower Type	Accessible Tiers	Loan Range	Use Case
Individual / End User	Local Bank only	0.01–10 ETH	Personal, micro-enterprise
Small Business / SME	Local Bank, National Bank	1–100 ETH	Working capital, equipment
Large Corporate	National Bank, World Bank	50–10,000 ETH	Infrastructure, large projects
Institutional / Sovereign	World Bank only	1,000+ ETH	Development programs

Local Banks are used by the end users and retail borrowers to gain access to the system, maintaining the hierarchical intermediation model. Greater levels can be accessed directly by large corporate accounts and institutional borrowers, but are subject to rigorous verification (on-chain credit history with off-chain documentation) and a borrower type designation in the smart contract that determines tier access permissions.

This multi-way lending scheme—downward capital allocation, same-tier interbank lending, upward repatriation of surpluses, and tiered access by borrowers—generates a more detailed model of the banking flows of the real world in the decentralized architecture. The current prototype fully implements the downward capital distribution and tiered borrower access components; same-tier interbank lending and upward surplus repatriation are architecturally designed and will be implemented in subsequent development phases.

1.8 Methodology in Brief

Our approach was a lightweight Agile/Scrum strategy consisting of three sprints in an 8-week development window. Sprint 1 creates smart contract, frontend framework, wallet integration, and database schema. Sprint 2 provides lending services (loan applications, approvals, installments, borrowing limits), chat system, income verification, and bank registration. Sprint 3 incorporates AI/ML fraud detection, risk dashboard, security audit, and documentation. Solidity 0.8.20 is used in development of smart contracts, React 18 with TypeScript on the frontend, FastAPI on the backend, and scikit-learn on the ML models. On Polygon and Ethereum, we deploy at zero real-cryptocurrency cost.

1.9 Scopes and Challenges

Scope. This prototype is restricted to publicly available testnets, i.e. no actual cryptocurrency is involved in the system. It encompasses basic features including four-tier lending architecture, loan request and approval process, installment based repayment, calculation of borrowing limits, communication between borrowers and banks and verification of income. The system also uses Random Forest with SHAP to explain explainability in fraud detection and Isolation Forest to detect anomalies. An additional feature of dual-currency is also introduced to enhance the current banking infrastructure. The prototype, however, does not cover mainnet deployment, fiat on/off-ramp integration, automated KYC/AML compliance or production-scale stress testing.

Challenges. There are some issues that come with this system. The unlabeled nature of DeFi lending fraud data is one of those problems, as it limits the successful training of the model and might necessitate the application of synthetic data or transfer learning methods. The other issue is regulatory uncertainty, since crypto lending may have different jurisdictional restrictions; and this risk can be partially reduced by working the testnets only. The sensitivity of gas costs is an issue as well, as intricate on-chain programs can be costly; to solve this, AI/ML work requiring computation is performed off-chain. Additionally, model interpretability is essential to regulatory compliance, especially in describing the process of lending, which is obtained by SHAP-based feature attribution. Last but not the least, economic sustainability should be taken into account, because the platform should be able to earn enough income, like interest revenues, to cover gas expenses at all four levels. The following Chapter 5 discusses this balance in greater depth.

1.9.1 Economic Sustainability: Interest Revenue Versus Gas Costs

The most important problem facing any blockchain-based lending system is whether the revenue earned by the lending activities will be in a sustainable position to cover the total gas expenses of the transactions on-chain. Every level in the hierarchy will have deposits, loan applications, approvals, disbursements, repayment and installment processing gas fees. One complicated smart contract interaction can cost between $5 and

\$50 on Ethereum mainnet, based on network congestion, but around \$0.001 to \$0.01 in Polygon PoS. At an average retail loan of 10 ETH (\$20,000 at \$2,000/ETH) with an 8% APR, the borrower pays \$1,600 in annual interest. With 10–15 on-chain transactions per loan lifecycle (request, approval, disbursement, 12 monthly installments), the overall gas costs on Polygon are under \$0.15—less than 0.01% of the interest obtained. The platform thus provides a sustainable margin on Layer 2 networks, but mainnet implementation would need to consider gas optimisation or batched transaction processing to ensure it would be viable to serve micro-loans.

1.9.2 Resistance to Institutional Capture

When the Crypto World Bank gains a mass following, the real question would be whether big capitalists and established financial institutions, who already dominate the traditional banking system, are able to migrate to the new system and repeat the established trends of financial concentration. This risk is mitigated by the platform in a variety of architectural ways:

- **Visible reserve ratios:** In contrast to traditional banks, where reserve adequacy is reported and audited quarterly, on-chain reserves are publicly verifiable in real time. There is no way a given institution can hide its real financial status to have unfair advantage.
- **Algorithmic interest rates:** Interest rates are determined by the smart contract parameters, not by opaque internal pricing committees, and thus no single participant can capture the entire pool of capital.
- **Immutable audit trails:** Every transaction, approval, and governance decision is permanently recorded on-chain, making regulatory capture and corruption detectable by any participant.
- **Open-source governance:** The smart contract code is publicly auditable, preventing hidden backdoors or preferential treatment encoded at the protocol level.
- **Tiered access with caps:** Borrowing limits and tier access rules are enforced programmatically, preventing any single entity from monopolizing the capital pool.

Although no system can entirely deter wealth concentration in a free market, the detectability and programmability characteristics of blockchain infrastructure make exploitative behavior prohibitively expensive and highly observable by comparison to opaque intermediaries.

1.9.3 Cryptocurrency Supply Constraints and Scalability

Among the most fundamental questions regarding the use of fixed-supply cryptocurrencies (e.g. the 21 million limit of Bitcoin or the deflationary issue of Ethereum after the Merge) that serve as denomination of a global lending system is what would happen when the demand of credit exceeds the supply of underlying cryptocurrency? Three design considerations are used to address this concern:

1. **Stablecoin support (planned):** The platform will have ERC-20 stablecoin support (USDC, USDT, DAI) built in, which are backed by fiat currencies and can be minted on demand—elastic supply with no on-chain transparency dumping.
2. **Multi-chain and Layer 2 deployment:** Multi-chain interoperability enables the platform to operate on multiple networks simultaneously, sharing the demand among different token ecosystems and eliminating bottlenecks in supplying individual chains.
3. **Fractional reserve design:** The hierarchical lending model is a hierarchical lending system, which is a fractional reserve system, in the sense that all the ETH deposited in the Tier 1 level can be lent out many times over via the multiplier effect, similar to a conventional banking system. The total lending capacity of the system is more than the gross amount of tokens supplied by a factor which is computed by the reserve ratio of each stage.

The fixed amount of the underlying cryptocurrency is not a weakness, but an opportunity: it removes the monetary discretionary growth that destabilizes buying power in traditional fiat economies. To scale, the system uses stablecoins and multichain implementation instead of printing money—the deflationary financial discipline that differentiates decentralized finance and central banking.

1.10 Market Analysis and Partnership Ecosystem

1.10.1 Market Sizing

Table 1.2: Market segments.

Segment	Description	Estimated Scale
Total Addressable Market (TAM)	Global DeFi lending ($55B+ TVL [13]); cross-border remittances ($860B [26]); SME gap ($4.5T [20])	$55B–5T+
Serviceable Addressable Market (SAM)	Institutional and semi-institutional lending requiring hierarchical structures	$5–15B
Serviceable Obtainable Market (SOM)	Pilot deployments in regulatory sandboxes, academic prototypes, NGO-backed microfinance	$50–200M

Sources: *(a) TAM: DefiLlama [13] reports DeFi lending TVL exceeding $55B; World Bank [26] values global remittances at $860B; IFC [20] estimates the MSME financing gap at $4.5T; (b) SAM and SOM: project-derived estimates based on industry segmentation [18].*

The market for **transparent, hierarchy-preserving decentralized lending** is presently unaddressed by existing DeFi protocols, which universally adopt flat, pool-based architectures. This represents a significant whitespace opportunity.

1.10.2 Target Customer Segment

The Crypto World Bank targets the **retail customer segment** — individual borrowers and small businesses seeking transparent, accessible crypto-based lending services.

Table 1.3: Target customer segment profile.

Characteristic	Description
Primary Users	Individual retail borrowers seeking personal or small business loans
Geographic Focus	Developing economies with limited traditional banking access (e.g., Bangladesh, Southeast Asia, Sub-Saharan Africa)
Loan Size Range	Micro to mid-range: 0.1 ETH – 500 ETH equivalent (~$200 – $1,000,000 at current rates)
User Profile	Digitally literate individuals with cryptocurrency wallet access; small business owners; gig-economy freelancers
Key Pain Points	High interest rates from informal lenders; lack of credit history in traditional systems; exclusion from banking due to documentation barriers

Why Retail: The retail segment represents the largest underserved population in developing economies. According to the World Bank's Global Findex Database, approximately 1.4 billion adults remain unbanked, with the majority concentrated in developing countries. By targeting retail customers, the platform maximizes financial inclusion impact while generating the transaction volume necessary to sustain the lending ecosystem. The hierarchical banking model (World Bank → National → Local → Borrower) mirrors how traditional microfinance institutions reach retail customers in underserved regions, but with blockchain-enforced transparency and AI-enhanced risk assessment.

1.10.3 Partner Ecosystem

Table 1.4: Partner categories and roles.

Partner Category	Functional Role	Blockchain-Mediated Incentive
Financial Regulators	Regulatory sandbox approval; compliance oversight	Reduced enforcement cost through on-chain transparency and audit trails
Banking Institutions	Network membership as National/Local Banks	Access to diversified global reserve; reduced inter-bank settlement friction
Payment Gateway Providers	Fiat-to-crypto on-ramp and off-ramp services	Volume-based transaction fees; expanded market reach
Academic & Research Institutions	Validation of AI/ML models; publication of research findings	Access to anonymized datasets; collaborative research opportunities
Non-Governmental Organizations	Pilot deployment; field testing with underserved borrower populations	Transparent, low-friction credit access for beneficiaries

1.10.4 Incentive Alignment Through the Blockchain Platform

The system manages and enforces partner incentives directly on the blockchain, making the process more transparent and reliable:

- **Immutable repayment records** serve as an on-chain reputation system through permanent repayment records, allowing credit decisions to be based on actual data without relying on external credit bureaus.
- **Transparent reserve verification** enables on-chain reserve verification, so all participants can independently confirm that allocated funds are being used as intended.
- **Programmable fee structures** (with potential for future expansion) allow

transaction fees to be distributed fairly among network participants according to their roles and contributions.

Chapter 2

Literature Review

2.1 Preliminaries

This part defines important terminology and concepts which are employed in the literature review and the project design.

Decentralized Finance (DeFi). DeFi means financial services that are based on blockchain places that do not use conventional intermediaries. Lending protocols allow borrowing and lending of assets via smart contracts, at interest rates that are usually algorithmically calculated or market utilized.

Smart contracts. Self-sovereign programs run on a blockchain that execute when certain preset conditions are fulfilled. Smart contracts are used to manage deposits in lending, loan applications, loans, loan repayments, and interest calculations without human intermediaries.

Over-collateralization. Borrowing model in which borrowers are required to give security at a higher amount than the loan amount. This is the model that prevails in DeFi (e.g., Aave, Compound) because it lacks credit scoring; it does not include under-collateralized lending of the conventional finance.

Explainable AI (XAI). Methods that cause machine learning model predictions interpretable to humans. SHAP and Local Interpretable Model-agnostic Explanations (LIME) are leading techniques of assigning importance of features to individual forecasts, which are important in the regulatory compliance of loans.

Proof of Stake (PoS). A network security mechanism of validators through postage of tokens; applied by Ethereum (after the merge) and Polygon. PoS is more efficient than Proof of Work and allows the block finality to be faster.

Central Bank Digital Currency (CBDC). A computerized variant of fiat currency of a nation, centrally issued. CBDCs incorporate leveled distribution models that are parallel to the multi-tier design that was discussed in this project.

2.2 Review of Existing Research

The idea of the Crypto World Bank is designed based on the examination of peer-reviewed materials, institutional reports, and industry figures covering more than one sphere of direct interest to our architecture: blockchain machine learning architecture, decentralized lending protocol architecture, security, explainable AI in finance, blockchain for financial inclusion, smart contract security, correspondent banking economics, monetary policy distribution effects, and real world asset tokenization.

2.2.1 Decentralized Lending and DeFi Protocol Design

In their SoK paper, Werner et al. [1] systematize DeFi protocol design and found that current lending markets are homogenous, pool based, and over-collateralized, and they do not exist institutional hierarchy—a gap which is directly addressed by this project. Their taxonomy of DeFi protocols indicates that there is no current system modeling that can represent multi-tier capital flow in a similar manner to development finance, ensuring the innovativeness of our four-level approach.

Bastankhah et al. [2] present a data-driven DeFi lending protocol that is adaptive dual fast/slow control architecture which proves the dynamic interest rate adjustment compares very well with the static utilization curves employed by Aave and Compound. Their experience confirms that it is a viable method to optimize the algorithmic lending parameters, which is justified by work we go through our adjustable levels of interest rates, through the four levels of the hierarchy.

Li et al. [46] introduce a multi-chain lending model, which allocates lending activities on various blockchain networks in order to enhance throughput and lower single-chain congestion. Their cross-chain settlement system certifies technical feasibility of executing lending protocols in heterogeneous blockchain settings, directly promoting our intended multi-chain implementation approach on Ethereum and Polygon.

Xu et al. [47] come up with a DeFi lending protocol evaluation system that quantifies protocol risk by measuring such metrics as the stability of utilization ratio, liquidation efficiency, and interest rate responsiveness. Their assessment system gives them a flexible methodology, which can be implemented to evaluate the performance of our hierarchical lending layers versus the current flat-pool protocols.

Sharma et al. [48] introduce a peer-to-peer lending framework that is powered by blockchain that removes intermediary overhead by the use of smart contract-mediated loan origination and repayment. Although they are still single-tier in architecture, the cost of their gas analysis on-chain credentials and management of identity through optimization strategies and borrowers guides our strategy in reducing transaction costs in the four level hierarchy.

2.2.2 Machine Learning for Blockchain Security

Palaiokrassas et al. [3] use machine learning on multichain DeFi fraud over 54 million transactions in 23 protocols which show a demonstration that the behavioral characteristics of DeFi enhance Neural Network and XGBoost classifiers to F1-scores of 0.76–0.85 versus 0.08 using transactional features only. Their finding that the behavioral features are superior to raw transaction data directly informed our feature engineering method for the random forest fraud detection model.

The algorithm of unsupervised anomaly detection presented by Liu et al. [6] is the Isolation Forest algorithm. The algorithm identifies anomalies using recursive partitioning of data, which assigns shorter distances to outliers. We used Isolation Forest as the second detection model when there are limited labeled fraud data to analyze wallet behavior, it is possible to detect new patterns of attacks not pre-labeled.

Hassan et al. [49] introduce blockchain and machine learning to detect fraud with the help of a privacy-protecting and adjustment-flexible incentive-based method. Their framework trains ML classifiers on encrypted transaction characteristics via federated learning, whereby the operators of the detection model never get to see the individual transaction data. Our future extensions of our AI/ML layer involve the privacy-preserving paradigm, in which the histories of borrower transactions should be confidential and at the same time allow system-wide fraud pattern recognition.

2.2.3 Explainable AI in Financial Decision Systems

Direct comparison of LIME and SHAP on loan approval is given by Adom et al. [4], SHAP offers more consistent and in-depth feature attributions, which are found to be deeper via Shapley values whereas LIME provides quicker running time but less sure descriptions across repeated evaluations. We have used their comparison to make the decision to implement SHAP since the main explainability technique of loan risk measurements, trading off interpretability depth with regulatory compliance requirements.

Gupta et al. [56] use interpretable models that are based on SHAP on credit default assessment proving that SHAP feature attributions on Gradient Boosting and Random Forest classifiers are able to score over 0.92 with complete interpretability of individual predictions. Their approach to causing per-borrower risk explanations tells our intended credit risk dashboard, where the approvers of loans look at SHAP waterfall plots before lending decisions are made.

2.2.4 Blockchain for Financial Inclusion

Tan [5] constructs a model of the International Monetary Fund (IMF) according to which CBDCs in developing countries can bank large unbanked populations using two-tier distribution model (central bank → commercial banks → users) that directly parallels our four-tier hierarchy. The paper shows that the distributed infrastructure of digital currency can prove to be effective by use of available institutional means, can access those populations not covered by traditional banking—facilitating the mission of the Crypto World Bank, which is transparent and programmable lending to underserved populations.

The article by Alam et al. [54] examines how blockchain technology can be used in the microcredit industry particularly in Bangladesh, it can be proven that smart contract-based loan management is capable of minimizing the administrative overhead by up to 60% as compared to manual microfinance processes. They have analyzed the Bangladeshi financial environment where about 40% of adult population do not have access to formal banking—directly confirms the geographic and demographic attractiveness of the target market of the Crypto World Bank.

The article by Islam et al. [53] examines the design requirements and challenges of Central Bank Digital Currencies, there are two levels of retail CBDC model (central bank to commercial banks to end users) as the distribution architecture of choice. Their analysis of interoperability challenges between CBDC systems and existing payment infrastructure informs our design of the dual-currency facility that bridges fiat and cryptocurrency within participating banks.

2.2.5 Smart Contract Security and Governance

Atzei et al. [7] list Ethereum smart contract attack vectors such as reentrancy, access control vulnerabilities, integer overflow, and access control vulnerabilities. They were directly informed by their taxonomy our implementation of the ReentrancyGuard by OpenZeppelin, Solidity 0.8.20 inbuilt overflow protection, and role-based access control modifiers. On their advice we took up their suggested defense-in-depth strategy of combining security primitives with formal verification with planning using static analysis (Slither) and symbolic execution (Mythril).

The article by Wang et al. [50] introduces ContractWard which is an automated vulnerability detection system that uses machine learning classifiers (Random Forest, SVM, k-NN) to classify six types of smart contract vulnerabilities based on the features of bytecodes. ContractWard scores F1-reentrancy and F1-access control with 0.96 and 0.93 respectively, proving that security auditing using ML can be used to supplement traditional static analysis tools. Our proposed security verification pipeline will utilize the use of machine learning-based static analysis tools.

Liao et al. [51] present SoliAudit, that is an integration of machine learning classification and vulnerability assessment of smart contracts fuzz testing. Their hybrid approach—using ML to first prioritize probable vulnerable code paths then target fuzz testing—reduces audit time by an estimated 70% of that spent on manual review. SoliAudit's methodology informs our intended security audit plan on the three-contract architecture.

So et al. [52] come up with VERISMART, an extremely accurate Ethereum safety verifier which employs constraint-based abstract interpretation to detect arithmetic overflow, division-by-zero and array out-of-bound errors. VERISMART achieves 98.4% accuracy on benchmark data, but has zero false positives, so it is a candidate tool to formally verify our smart contract invariants (e.g., reserve ratio maintenance, cascading interest rate limits).

2.2.6 AI Security Features for Blockchain Lending

In addition to the currently existing fraud detection and anomaly identification models in place in the platform, there are other security features that are based on AI that can enhance the Crypto World Bank's resilience to threats of change:

- **Graph Neural Network (GNN) transaction analysis:** GNN-based models are able to analyze the graph structure of transaction to identify coordinated rings of fraud—groups of wallets which conspire to rig the borrowing limits or make circular lending patterns. A study by Palaiokrassas et al. [3] proves that graph-based features are much superior to flat transaction features as an indicator of DeFi fraud detection.
- **Federated learning for cross-bank fraud intelligence:** Hassan et al. [49] demonstrate that federated learning allows the collaboration between several institutions to detect and train fraud models without distributing raw transaction information. Deploying federated learning of the Crypto World Bank's National and Local Banks would allow detecting threats system wide without compromising the privacy of the per-bank data.
- **Reinforcement learning for adaptive interest rates:** As a reinforcement-based learning algorithm, adaptive interest rates can be learned dynamically; RL agents can dynamically modify interest rate parameters to match the varying market conditions, utilization rates, and risk profiles, decreasing the lag of manual parameter governance.
- **Real-time smart contract monitoring:** ML-based monitoring agents (informed by ContractWard [50] and SoliAudit [51]) can continuously analyze incoming transactions for patterns matching known exploit signatures, triggering the platform's

pause mechanism before significant losses occur.

- **Natural language processing for income verification:** NLP models are able to extract and validate income information in uploaded documents, saving the bank officers the manual review load but not eliminating verification accuracy.

2.2.7 Gas Cost Optimization and Layer 2 Scalability

In DeFi, Tolmach et al. [55] examine the optimal approaches to minimization of gas fees, demonstrating that transaction batching, calldata compression, and tactical time of on-chain operations can save 30–50% of gas costs on Ethereum mainnet. Their findings test our architectural choice of deploying to Polygon PoS (where fees already are sub-cent) and inform optimization strategies of the contemplated Ethereum mainnet deployment. The research also finds out that complicated multi-step operations of DeFi (analogous to our four-tier loan lifecycle) benefit disproportionately from Layer 2 deployment due to the multiplicative savings in gas of sequential calls of contracts.

2.2.8 Correspondent Banking and Cross-Border Settlement

The Bank of International Settlement [21] and Financial Stability Board [33] have also written a lot about the structural inefficiencies of the correspondent banking network. In the traditional model, the banks are required to have pre-funded nostro accounts in every currency corridor, establishing a world pool of idle capital which can not be put into effective productive lending. SWIFT settlements and respondent banks involves routing of messages using MT103 payment directions, screening of compliance at all intermediates, and settlement time extensions to two–five business days. According to the World Bank Migration and Development Brief [26] it is stated that the cost of transferring $200 on cross-border is 6.49% on average, with Sub-Saharan African corridors averaging more than 8%. These results encourage our development of on-chain settlement with close to instant finality and sub-cent transaction costs on Layer 2 networks.

2.2.9 Monetary Policy Distribution and Financial Inequality

Recent empirical studies have reported the distributional impact of monetary policy on wealth inequality. A 2025 cross-country study spanning 49 nations (1999–2019) discovered that the relationship between central bank asset purchases programs and greater wealth inequality, whose outcomes are stronger in the distribution of wealth and long lasting [34] than income inequality effects. The Federal Reserve's own 2025 contractionary monetary analysis based on the U.S. metropolitan tax data confirmed that policies harm the income of low income employees [35]. These findings deliver economic incentive to the design principles of Crypto World Bank: transparent and monetary parameters (interest rates, reserve ratios, supply rules) that have been algorithmically determined represented in smart contracts as opposed to being established discretionally by central bank decisions. On-chain transparency of the platform makes sure that all participants follow the same interest rates and reserve conditions, which removes the informational asymmetry permitting the Cantillon Effect—the effect that early receivers of new money created benefit at the expense of subsequent recipients who will suffer greater prices [36].

2.2.10 Real-World Asset Tokenization and Institutional DeFi

The tokenization of real-world assets is an emerging convergence of traditional and decentralized finance. The Corda platform of R3 boasts of more than $17 billion in tokenized real-world assets on its permissioned ledger as of September 2025, with institutional participants including HSBC and Bank of America [37]. Centrifuge has deployed in eight blockchain networks of more than $1 billion in tokenized institutional fund products [38]. The World Bank is using blockchain on its own public chain side to track the disbursement of funds using its FundsChain program (based on Hyperledger Besu) which is flown on 13 projects in 10 countries and increases to 250 projects by mid-2026 [39]. These developments confirm the institutional interest of blockchain-based financial infrastructure and the viability of hierarchical fund distribution design models on distributed ledgers. This institutional adoption is continued in the Crypto World Bank's trajectory through carrying out not only fund tracking but total hierarchical lending system that has on-chain interest rates, reserves and credit assessment.

2.3 Literature Review Summary

The most important reviewed works with their methodologies, key findings, and relevance to our project are presented in Table 2.1, in the form of literature table suggested by Michigan State University Libraries [23].

Table 2.1: Literature review summary table.

Author / Year	Research Focus	Methodology	Key Findings	Relevance to Our Project
Werner et al. (2022) [1]	DeFi protocol systematization	SoK survey of 12+ DeFi protocol categories	Lending platforms are uniformly pool-based and over-collateralized; no institutional hierarchy exists	Identifies the core gap our four-tier architecture addresses
Bastankhah et al. (2023) [2]	Adaptive DeFi lending	Dual fast/slow control; simulation on historical data	Dynamic rate adjustment outperforms static utilization curves	Validates algorithmic lending parameter optimization

Continued on next page

Table 2.1: Literature review summary table (continued).

Author / Year	**Research Focus**	**Methodology**	**Key Findings**	**Relevance to Our Project**
Palaiokrassas et al. (2023) [3]	ML fraud detection in DeFi	XGBoost, NN classifiers on 54M+ multi-chain transactions	DeFi behavioral features yield F1: 0.76–0.85 vs. 0.08 with transaction features alone	Informs our feature engineering for Random Forest fraud model
Adom et al. (2022) [4]	XAI in loan approval	LIME vs. SHAP comparison on lending datasets	SHAP provides deeper, more consistent attributions; LIME is faster but less stable	Justifies SHAP as our primary explainability method
Tan (2023) [5]	CBDC and financial inclusion	IMF economic model for developing nations	Two-tier CBDC distribution banks large unbanked populations	Directly parallels our four-tier institutional hierarchy
Liu et al. (2008) [6]	Anomaly detection	Isolation Forest algorithm on synthetic and real datasets	Isolation via recursive partitioning; shorter paths for anomalies	Adopted as secondary unsupervised detection for wallet behavior
Atzei et al. (2017) [7]	Smart contract security	SoK of Ethereum attack vectors	Catalogs reentrancy, overflow, access control vulnerabilities	Directly informs our security primitives and planned formal verification

Continued on next page

Table 2.1: Literature review summary table (continued).

Author / Year	Research Focus	Methodology	Key Findings	Relevance to Our Project
BIS/FSB [21][33]	Cross-border settlement	Analysis of correspondent banking infrastructure	Settlement takes 2–5 days; avg. cost $42/transfer; capital trapped in nostro accounts	Motivates on-chain settlement with near-instant finality
Beyer et al. (2025) [34]	Monetary policy and inequality	Cross-country analysis, 49 nations, 1999–2019	QE increases wealth inequality; effects more persistent than income effects	Motivates transparent, algorithmic monetary parameters
World Bank (2025) [39]	Blockchain for development finance	FundsChain pilot on Hyperledger Besu, 13 projects, 10 countries	Blockchain tracking reduces reporting from months to minutes	Validates institutional appetite for blockchain-based fund distribution
Li et al. (2024) [46]	Multi-chain DeFi lending	Cross-chain lending model design and implementation	Multi-chain distribution improves throughput and reduces congestion	Supports our planned multi-chain deployment strategy
Xu et al. (2023) [47]	DeFi lending protocol evaluation	Quantitative risk metrics for lending protocol assessment	Utilization ratio, liquidation efficiency, and rate responsiveness as key metrics	Provides benchmark methodology for evaluating our hierarchical tiers

Continued on next page

Table 2.1: Literature review summary table (continued).

Author / Year	Research Focus	Methodology	Key Findings	Relevance to Our Project
Sharma et al. (2021) [48]	Blockchain P2P lending	Smart contract-mediated loan origination framework	Eliminates intermediary overhead; gas cost optimization analysis	Informs gas cost minimization across four-tier hierarchy
Hassan et al. (2022) [49]	Privacy-preserving fraud detection	Federated learning on encrypted blockchain features	Privacy-preserving ML achieves comparable accuracy to centralized models	Informs future privacy-preserving fraud detection extensions
Wang et al. (2020) [50]	Smart contract vulnerability detection	ML classifiers on bytecode features	F1: 0.96 for reentrancy, 0.93 for access control detection	ML-based auditing complements our planned security verification
Liao et al. (2019) [51]	Smart contract audit automation	Hybrid ML classification + fuzz testing	Reduces audit time by 70% vs. manual review	Informs our security audit strategy for three-contract architecture
So et al. (2020) [52]	Smart contract formal verification	Constraint-based abstract interpretation	98.4% precision, zero false positives on benchmarks	Candidate tool for verifying smart contract invariants
Islam et al. (2024) [53]	CBDC design requirements	Analysis of retail CBDC distribution architecture	Two-tier model preferred; interoperability challenges identified	Informs dual-currency facility design

Continued on next page

Table 2.1: Literature review summary table (continued).

Author / Year	Research Focus	Methodology	Key Findings	Relevance to Our Project
Alam et al. (2021) [54]	Blockchain micro-credit in Bangladesh	Smart contract-based microfinance management	Reduces admin overhead by 60%; validates Bangladesh market	Directly validates our target geographic and demographic market
Tolmach et al. (2024) [55]	DeFi gas fee optimization	Transaction batching and calldata compression analysis	Gas savings of 30–50% through optimization strategies	Validates Layer 2 deployment and informs mainnet optimization
Gupta et al. (2024) [56]	SHAP-based credit default models	SHAP on Random Forest and Gradient Boosting classifiers	AUC above 0.92 with full interpretability	Informs per-borrower risk explanation methodology

2.4 Summary of Key Findings

The following summarized findings are obtained in the literature review and are directly informative to our design:

- **DeFi lending is structurally flat.** Available protocols (Aave, Compound, MakerDAO) operate pool-based or over-collateralized models with total value locked exceeding $55 billion [13]. Institutional hierarchy has no comparable protocol models to development finance. The research by CBDC [5] ascertains that tiered distribution is workable and promotes financial inclusion.
- **Correspondent banking is inefficient in structure.** Cross-border settlement spends two to five days at average costs of $42 per transaction [25]. The system traps important capital lying idle in nostro/vostro accounts [24], and remittance fees are estimated to consume between $48–56 billion a year [26]. On-chain

settlement on Layer 2 networks is known to cut the latency (to seconds) and the cost (to sub-cent levels) down.

- **ML enhances fraud detection in blockchain.** DeFi-specific behavioral features significantly work well as compared to transactional features (F1: 0.76–0.85 vs. 0.08 [3]). The Isolation Forest [6] allows unsupervised detection in which labeled data is scarce.
- **SHAP is explainable by regulators.** SHAP offers greater consistency and is better than the LIME in loan approval systems [4] in meeting prudential demands on open-minded automated decision-making.
- **Layered defense is needed in smart contract security.** Reentrancy, overflow, and vulnerabilities to access control are well documented [7]. ML-based vulnerability detection achieves F1-scores of above 0.93 [50], and hybrid ML-fuzz methods reduce audit time by 70% [51]. Verification tools are formal, i.e., VERISMART achieves 98.4% precision [52]. Automated primitives used with OpenZeppelin are recommended for production deployments to be verified.
- **Monetary policy leads to distributional inequality.** Cross-country evidence [34] and Federal Reserve research [35] confirm that quantitative easing disproportionately favors asset holders, and workers of lower income cover the expenses of both inflation and consequent tightening. Transparent, algorithmic monetary parameters can decrease this informational asymmetry.
- **The use of blockchains in institutions is gaining momentum.** The World Bank's FundsChain [39], tokenized assets of R3 Corda worth $17 billion [37], and JPMorgan Kinexys settling multi-billion dollars per day [40] reflect the fact that financial institutions are actively implementing blockchain on settlement, funds tracking, and asset tokenization.
- **Blockchain allows financial inclusion.** Approximately 1.4 billion adults were not banked in any country around the world [14], and DeFi was found by the World Economic Forum as a leaping frog technology that allows peoples to avoid the banking systems infrastructure [41]. The MiniPay wallet of Celo has onboarded 14 million users in over 60 countries with less than cent transaction fees [42]. Blockchain-based microcredit in Bangladesh shows 60% decrease on administrative overhead [54].
- **Multi-chain lending enhances scalability.** Cross-chain lending models [46] and Layer 2 gas optimization strategies [55] prove the fact that the lending is distributed when operating on several networks, decreasing congestion and transaction costs by 30–50%.
- **ML can be used to detect cross-institutional fraud using privacy preserving ML.** Federated learning methods [49] are such that enable several lending institutions to cooperate and learn fraud detection models in a non-intrusive man-

ner, i.e. no individual transaction data is exposed, resolving the conflict between institution-wide security and data privacy.

The results explain why we have a four-tier architecture, AI/ML integration (Random Forest, Isolation Forest, SHAP), cross-tier lending structure, governance structure, and planned security verification pipeline.

Chapter 3

System Architecture and Design

3.1 High-Level Architecture

The system employs a **three-layer decentralized application architecture**:

PRESENTATION LAYER

React 18 · TypeScript · Material Design 3 · Wagmi · Viem

Modules: Dashboard, Deposit, Loan, Admin, Risk AI, QR

↓

SMART CONTRACT LAYER (EVM: Ethereum / Polygon)

World Bank Reserve Contract · National Bank Contract · Local Bank Contract

Operations: Reserve Mgmt, Hierarchical Lending, Loan Lifecycle, Access Control, Emergency Controls

↓

OFF-CHAIN SERVICES LAYER

Express.js (REST API) · **PostgreSQL** (Relational DB) · **FastAPI** (ML Inference Service)

AI/ML: Random Forest (Fraud) · Isolation Forest (Anomaly) · SHAP (Explainability)

Event Listener · Cache (Redis)

Figure 3.1 shows the component interactions across these three layers.

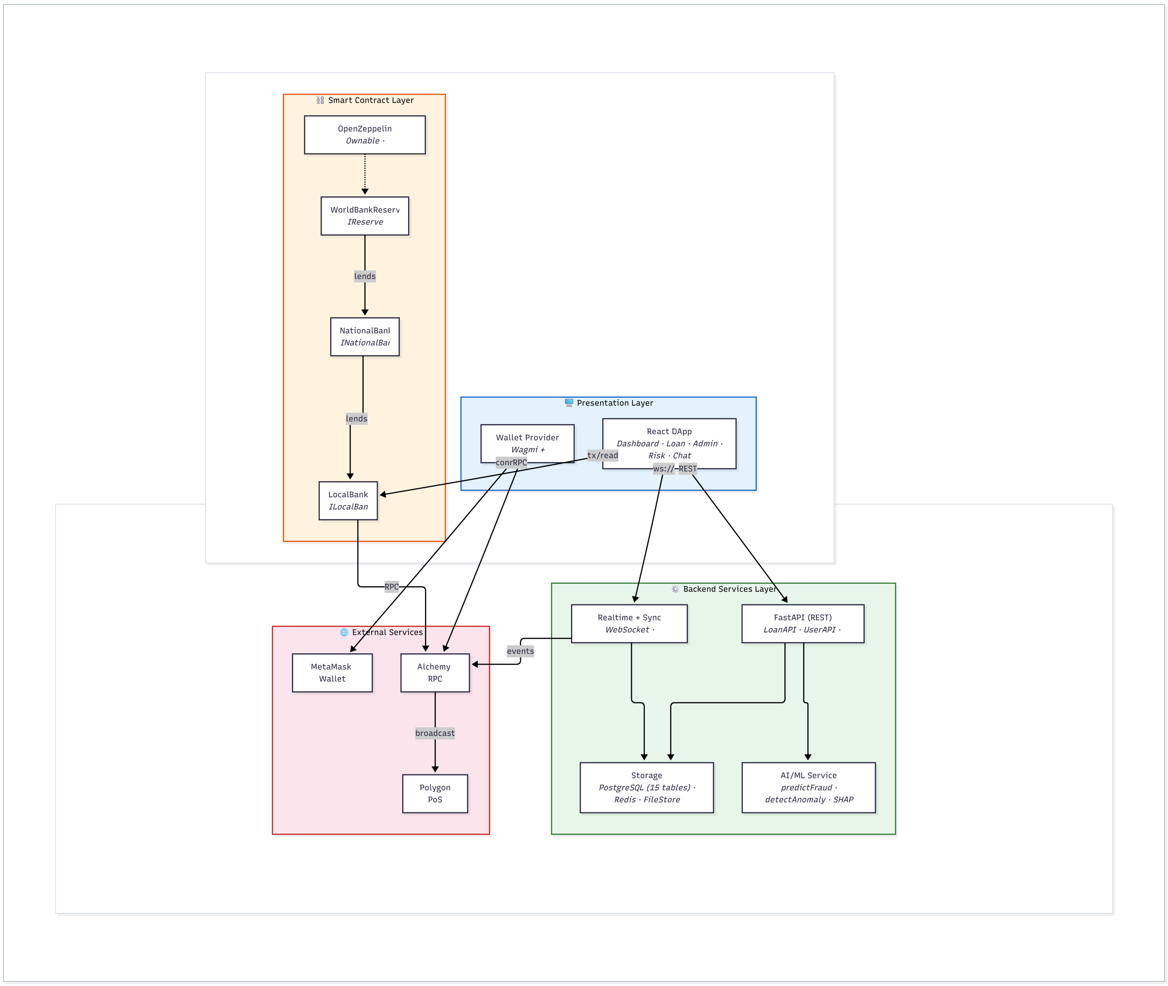

Figure 3.1: Component diagram showing interactions between the presentation layer, smart contract layer, off-chain backend services, and external systems.

3.2 Blockchain Platform Selection

PRESENTATION LAYER

React 18 · TypeScript · Material Design 3 · Wagmi · Viem

Modules: Dashboard, Deposit, Loan, Admin, Risk AI, QR

↓

SMART CONTRACT LAYER (EVM: Ethereum / Polygon)

World Bank Reserve Contract · National Bank Contract · Local Bank Contract

Operations: Reserve Mgmt, Hierarchical Lending, Loan Lifecycle, Access Control, Emergency Controls

↓

OFF-CHAIN SERVICES LAYER

Express.js (REST API) · **PostgreSQL** (Relational DB) · **FastAPI** (ML Inference Service)

AI/ML: Random Forest (Fraud) · Isolation Forest (Anomaly) · SHAP (Explainability)

Event Listener · Cache (Redis)

Table 3.1: Blockchain platform selection criteria and justification.

Criterion	Selection	Justification
Platform	Ethereum Virtual Machine (EVM)	Largest developer ecosystem; battle-tested security model; extensive tooling
Network	Polygon Amoy / Ethereum Sepolia (testnets)	Zero-cost deployment; production-equivalent behavior; free faucet access
Consensus	Proof-of-Stake (PoS) via Polygon validators	Energy-efficient; sub-2-second block finality; decentralized validator set
Smart contract language	Solidity 0.8.20	Industry standard; mature compiler with overflow protection; rich library ecosystem

3.2.1 Transaction Verification and Consensus

- **Polygon PoS:** A group of PoS validators that are not connected to each other checks transactions. It takes about two seconds to reach block finality. For extra security, checkpoints are sometimes sent to the Ethereum mainnet.
- **On prototype testnets:** Amoy and Sepolia use similar consensus models that don't cost any money, which lets them develop and test in small steps.

3.3 Data Model and Database Design

The relational database schema comprises 15 normalized entities in Third Normal Form (3NF). Figure 3.2 presents the core system graph showing entity relationships, Figure 3.3 presents the full Entity-Relationship Diagram (ERD), and Figure 3.4 presents the Enhanced Entity-Relationship (EER) diagram.

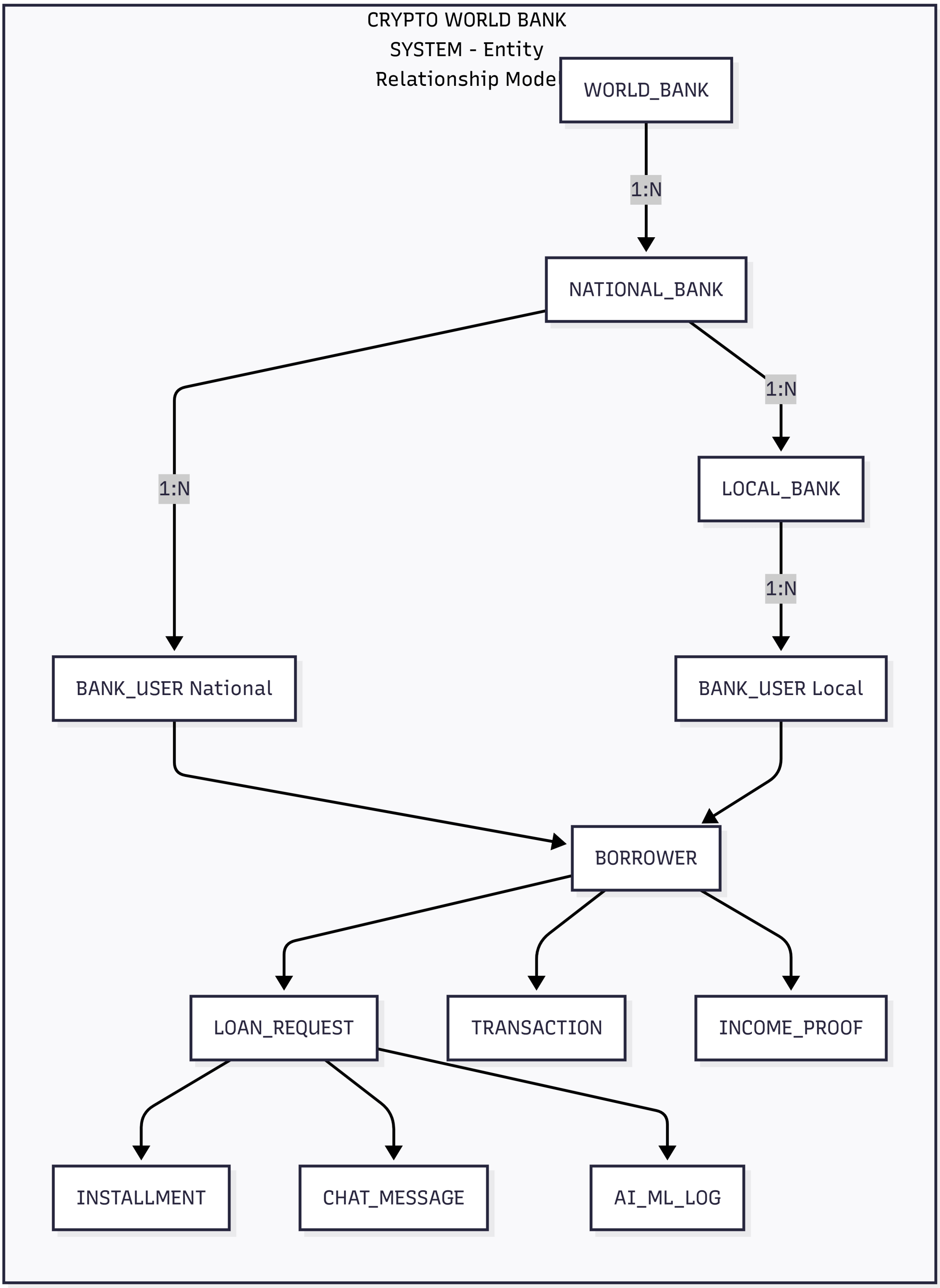

Figure 3.2: Core system graph showing the Crypto World Bank entity relationship model with hierarchical banking structure, lending lifecycle, communication, and security entities.

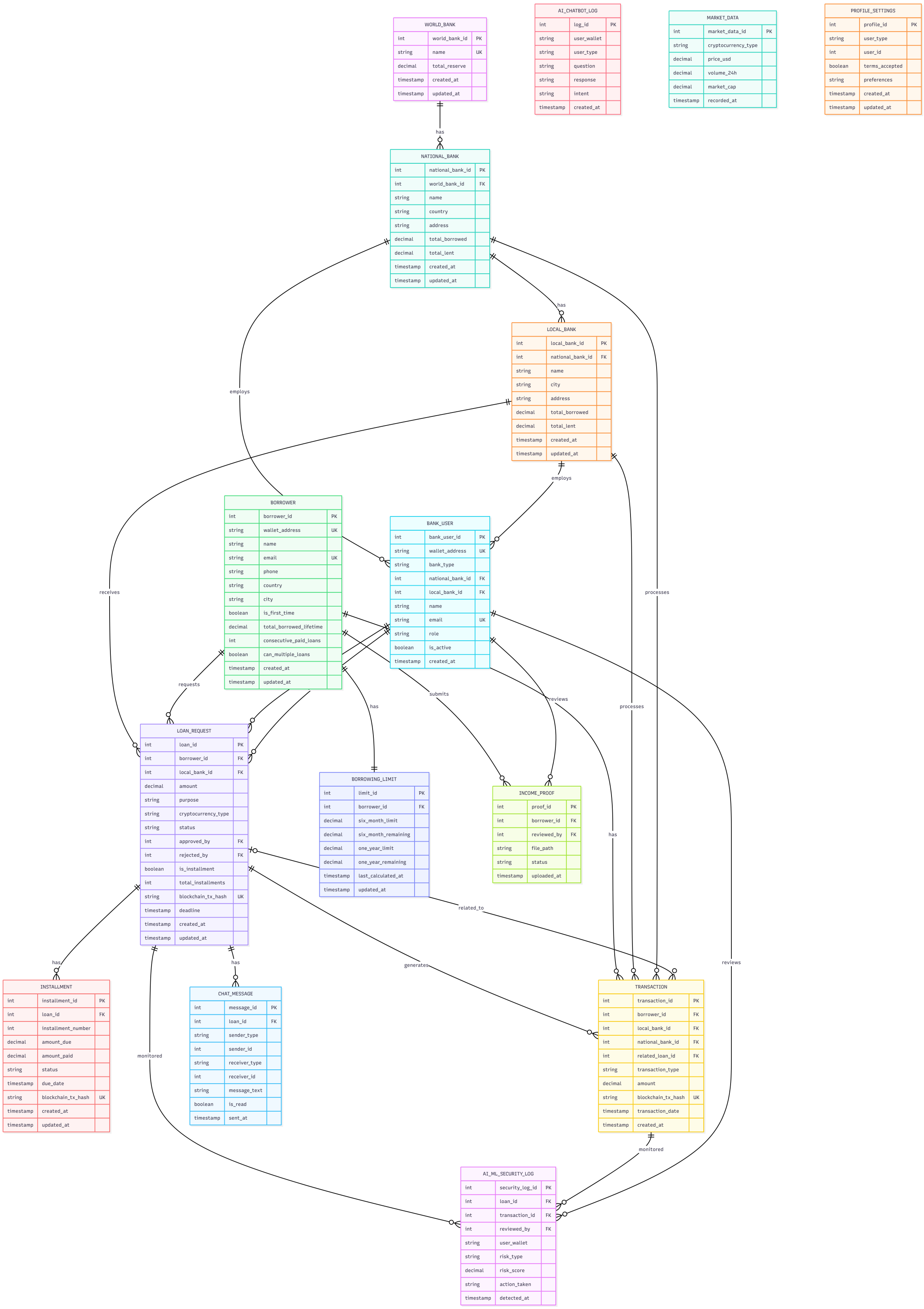

Figure 3.3: Entity-Relationship Diagram (ERD) for the Crypto World Bank database with relational symbols and entity attributes across all 15 tables.

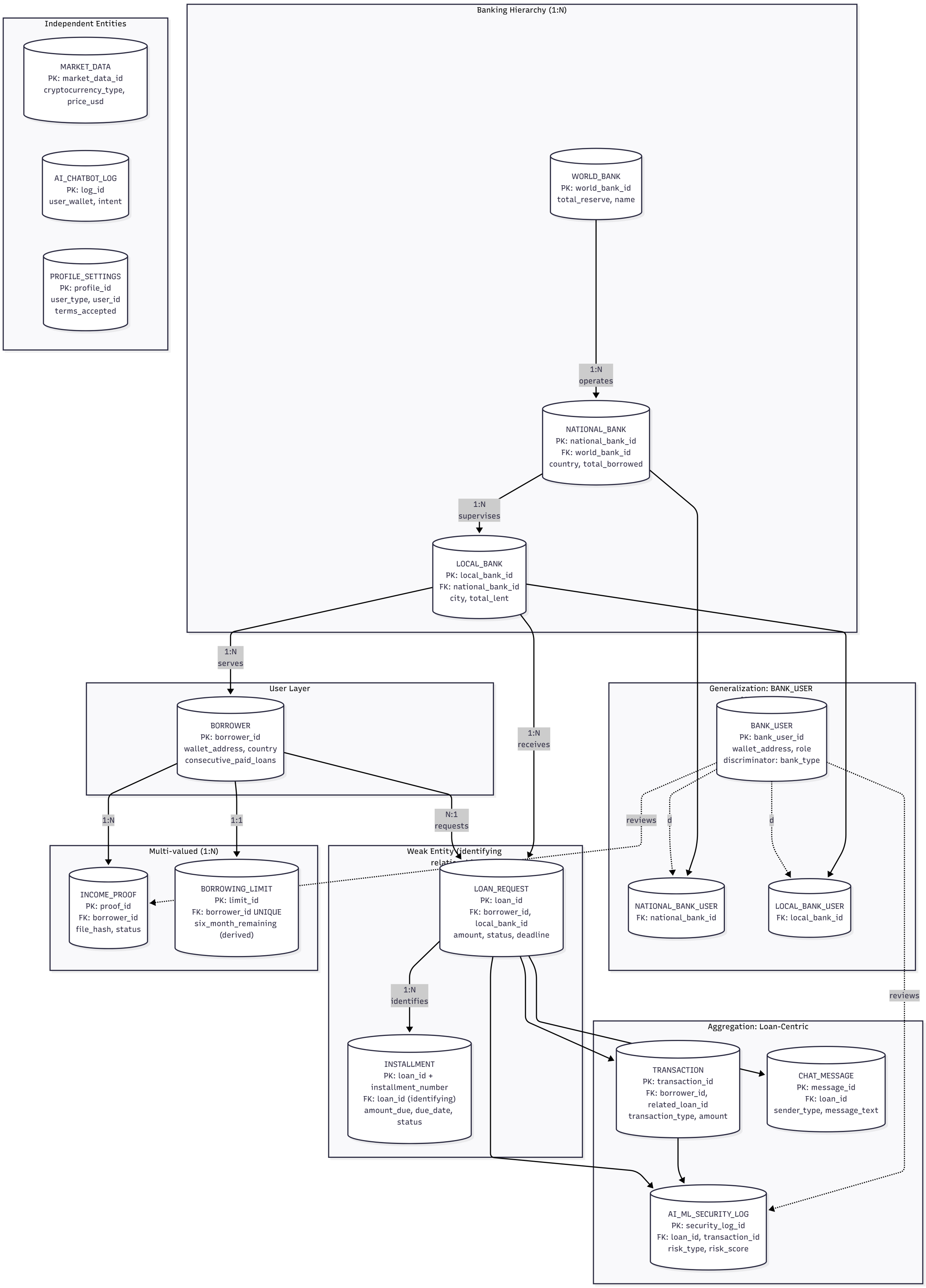

Figure 3.4: Enhanced Entity-Relationship (EER) diagram showing generalization/specialization, weak entities, aggregation, participation constraints, and cardinality for the complete data model.

3.3.1 Entity Summary

Table 3.2: Database entity summary (15 entities).

Entity	Role
WORLD_BANK	Top-level reserve holder; global lending parameters
NATIONAL_BANK	Country-level banks; borrow from World Bank
LOCAL_BANK	City-level banks; retail lending to borrowers
BANK_USER	Bank staff with role-based permissions (approve/reject loans)
BORROWER	End-users requesting and repaying loans
LOAN_REQUEST	Loan applications and full lifecycle tracking
INSTALLMENT	Installment payment records (weak entity dependent on LOAN_REQUEST)
TRANSACTION	Financial transaction records
BORROWING_LIMI'	Per-borrower limits with derived 6-month/1-year windows
INCOME_PROOF	Income verification documents (multi-valued per borrower)
CHAT_MESSAGE	Borrower–bank communication
AI_CHATBOT_LOG	AI chatbot interaction records
AI_ML_SECURITY_	Security and ML monitoring events
MARKET_DATA	Cryptocurrency price feeds
PROFILE_SETTING	User profile and platform preferences

3.3.2 EER Constructs Applied

Table 3.3: EER constructs applied in the Crypto World Bank data model.

EER Construct	Application
Generalization (disjoint)	BANK_USER generalizes to NATIONAL_BANK_USER and LOCAL_BANK_USER via `bank_type` discriminator with mutually exclusive foreign keys
Weak entity	INSTALLMENT is existence-dependent on LOAN_REQUEST with partial key `installment_number`
Aggregation	LOAN_REQUEST aggregates BORROWER + LOCAL_BANK; TRANSACTION, CHAT_MESSAGE, and AI_ML_SECURITY_LOG are components
Multi-valued attribute	INCOME_PROOF modeled as a separate entity (borrower can have multiple proofs)
Derived attribute	`six_month_remaining` and `one_year_remaining` in BORROWING_LIMIT are computed from TRANSACTION history
Composite key	INSTALLMENT uses (`loan_id`, `installment_number`)
Participation constraints	BORROWER has total participation in LOAN_REQUEST; INSTALLMENT has total participation when `is_installment = TRUE`
Cardinality	1:1 (BORROWER–BORROWING_LIMIT), 1:N (hierarchy, loan–installment), N:1 (loan–borrower)

3.3.3 Normalization

The schema has been normalized to Third Normal Form (3NF) and checked against Boyce-Codd Normal Form (BCNF):

- **1NF:** There are no repeating groups in any of the attributes. Separate entities store multi-valued attributes, such as proof of income.

- **2NF:** There are no partial dependencies. The full composite key (`loan_id`, `installment_number`) determines the non-key attributes of INSTALLMENT.
- **3NF:** No dependencies that go through other dependencies. Instead of storing redundant values like `total_borrowed` in bank entities, they are calculated at query time.
- **BCNF:** All determinants are candidate keys. A CHECK constraint on BANK_USER enforces that the generalization hierarchy's specializations are not the same.

3.3.4 Indexing Strategy

We use B-tree indexes (the default in PostgreSQL) for efficient retrieval on frequently queried columns:

Table 3.4: Database indexing strategy.

Table	Index	Purpose
LOAN_REQUEST	`idx_status_borrower`, `idx_status_bank`, `idx_deadline`	Filter pending loans; lookup by borrower or bank
TRANSACTION	`idx_type_date`, `idx_borrower_date`, `idx_date`	Borrowing limit queries (6-month/1-year windows)
BORROWER	`idx_wallet`, `idx_location`	Wallet address lookup; location-based queries
CHAT_MESSAGE	`idx_loan_sent`, `idx_sender`, `idx_receiver`	Chat history retrieval; unread message counts

B-tree indexes are chosen for their efficiency with range queries (e.g., transactions within the last 6 months); hash indexes would be suitable only for exact-match lookups.

3.3.5 Functional Dependencies

Table 3.5: Representative functional dependencies.

Relation	Functional Dependency	Notes
LOAN_REQUEST	`loan_id` → all attributes	Primary key determines row
LOAN_REQUEST	`borrower_id, local_bank_id` → `status, amount`, ...	One active request per borrower per bank
BORROWING_LI	`borrower_id` → `six_month_limit, one_year_limit`, ...	1:1 with BORROWER

3.3.6 Relational Integrity Constraints

Table 3.6: Relational integrity constraints.

Constraint Type	Examples
Primary Key	`world_bank_id`, `loan_id`, `borrower_id`, etc.
Foreign Key	`local_bank_id` in LOAN_REQUEST references LOCAL_BANK
UNIQUE	`wallet_address` in BORROWER; `blockchain_tx_hash` in LOAN_REQUEST
CHECK	BANK_USER: `(bank_type='national' AND national_bank_id IS NOT NULL) OR (bank_type='local' AND local_bank_id IS NOT NULL)`
NOT NULL	Core attributes: `name`, `wallet_address`, `status`

3.4 On-Chain and Off-Chain Data Partitioning

Table 3.7: Data partitioning between on-chain and off-chain storage.

Data Category	Storage	Rationale
Reserve balances, loan requests, approval/rejection events, repayment transactions	On-chain	Immutability, public auditability, trustless verification
User profiles, income verification documents, chat messages, AI/ML inference logs	Off-chain (database)	Data privacy, query flexibility, storage cost optimization
Borrowing limit computations	Off-chain with on-chain enforcement	Complex temporal aggregation; results committed as on-chain constraints
Cryptocurrency market data	Off-chain (cached)	High-frequency updates; external API dependency

3.5 Digital Identity System

- **Identity based on wallets:** Users log in with Ethereum wallet signatures (MetaMask, WalletConnect). You don't need a central place to store credentials.
- **Role binding:** In the smart contract permission system, wallet addresses are linked to hierarchical roles like Owner, National Bank, Local Bank, Approver, and Borrower.

3.6 System Modeling

This section presents the system analysis diagrams that model the platform's structure, behavior, and data flow.

3.6.1 Use Case Diagram

The system involves four primary actors—Borrower, Bank Approver, World Bank Admin, and National Bank—across 29 use cases. Figure 3.5 shows the complete use case diagram.

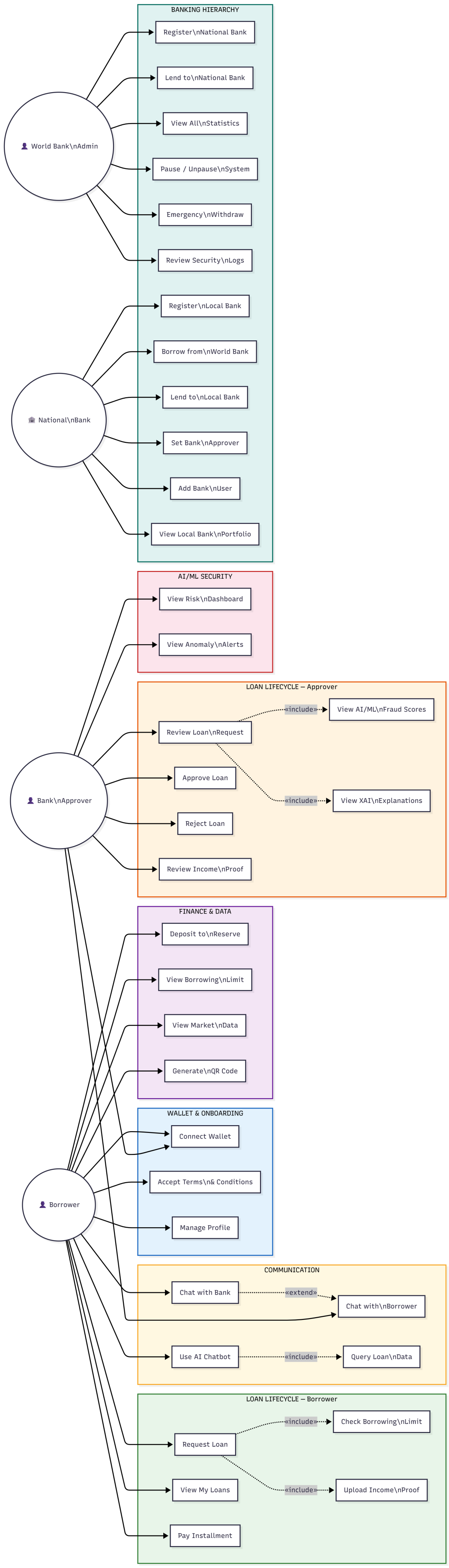

Figure 3.5: Usecase diagram

3.6.2 Activity Diagrams

Figures 3.6–3.12 present the activity diagrams for the platform's seven core operational flows.

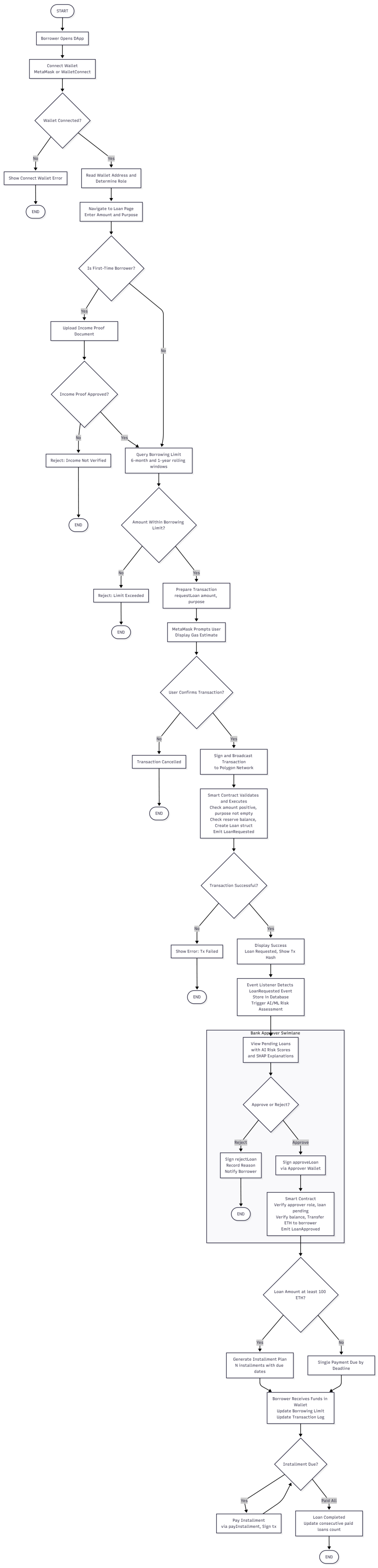

Figure 3.6: Activity Diagram - Loan Request to Repayment Flow

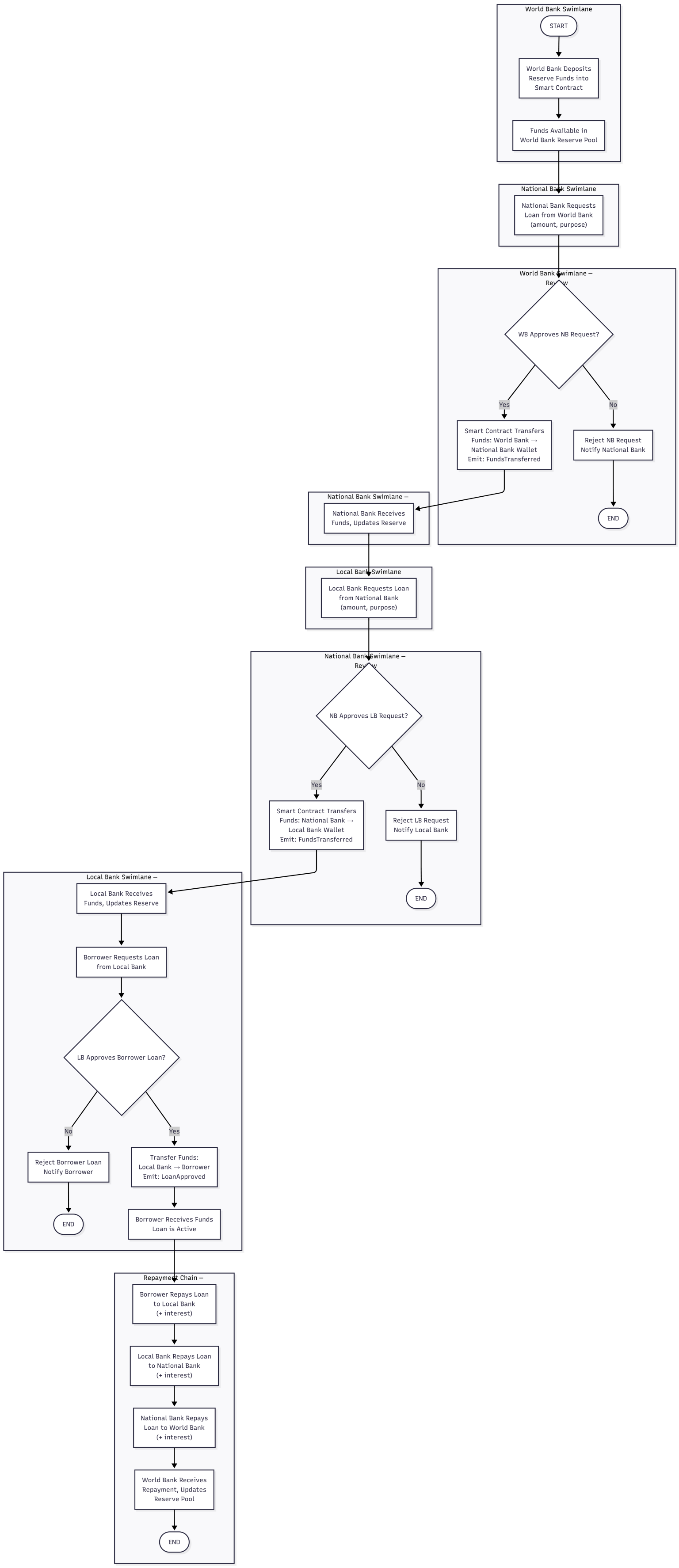

Figure 3.7: Activity Diagram Hierarchical Banking Flow

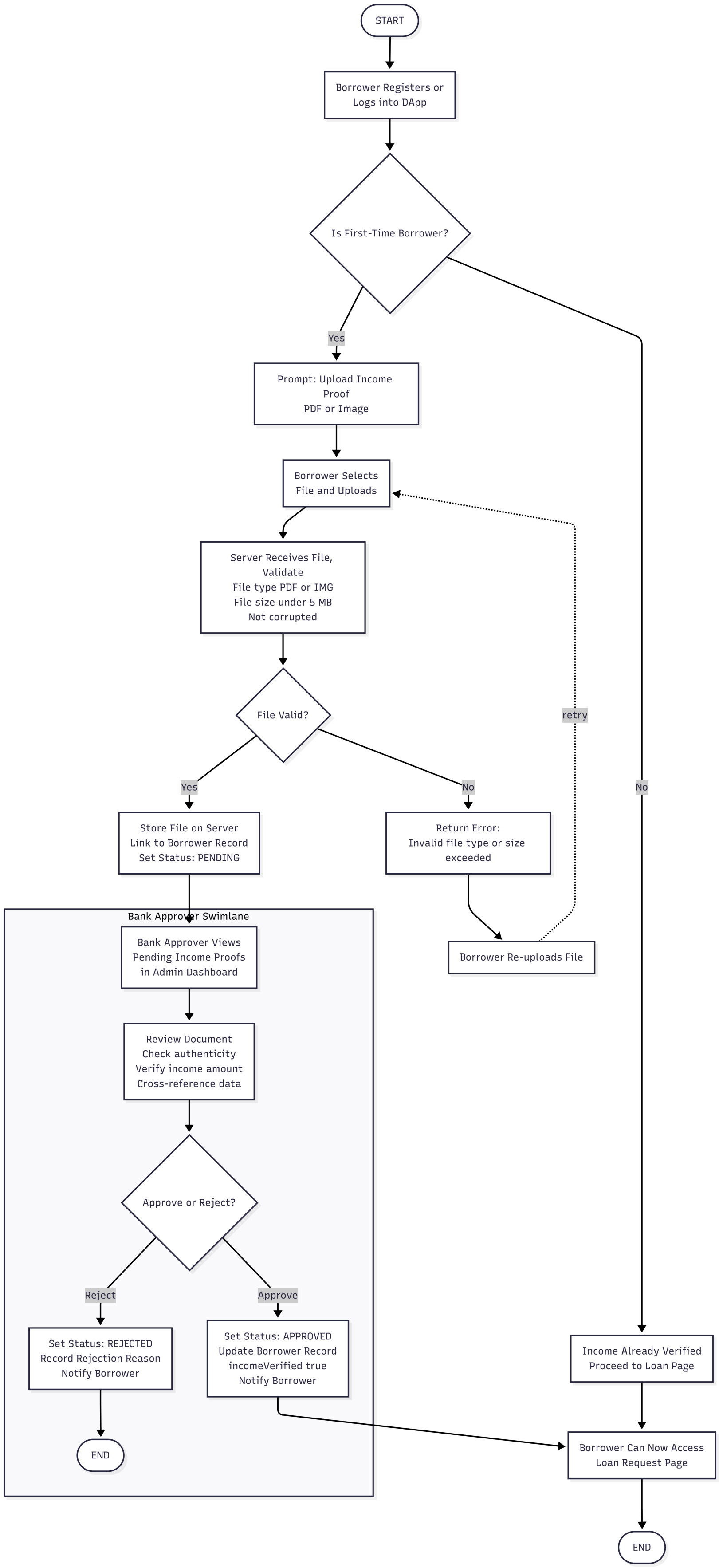

Figure 3.8: Activity Diagram Income Verification Flow

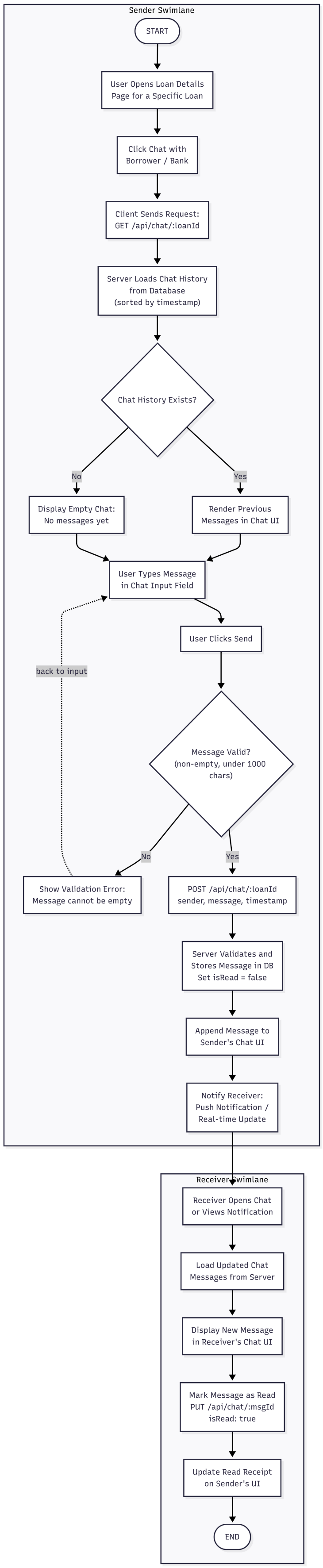

Figure 3.9: Activity Diagram Chat System Flow

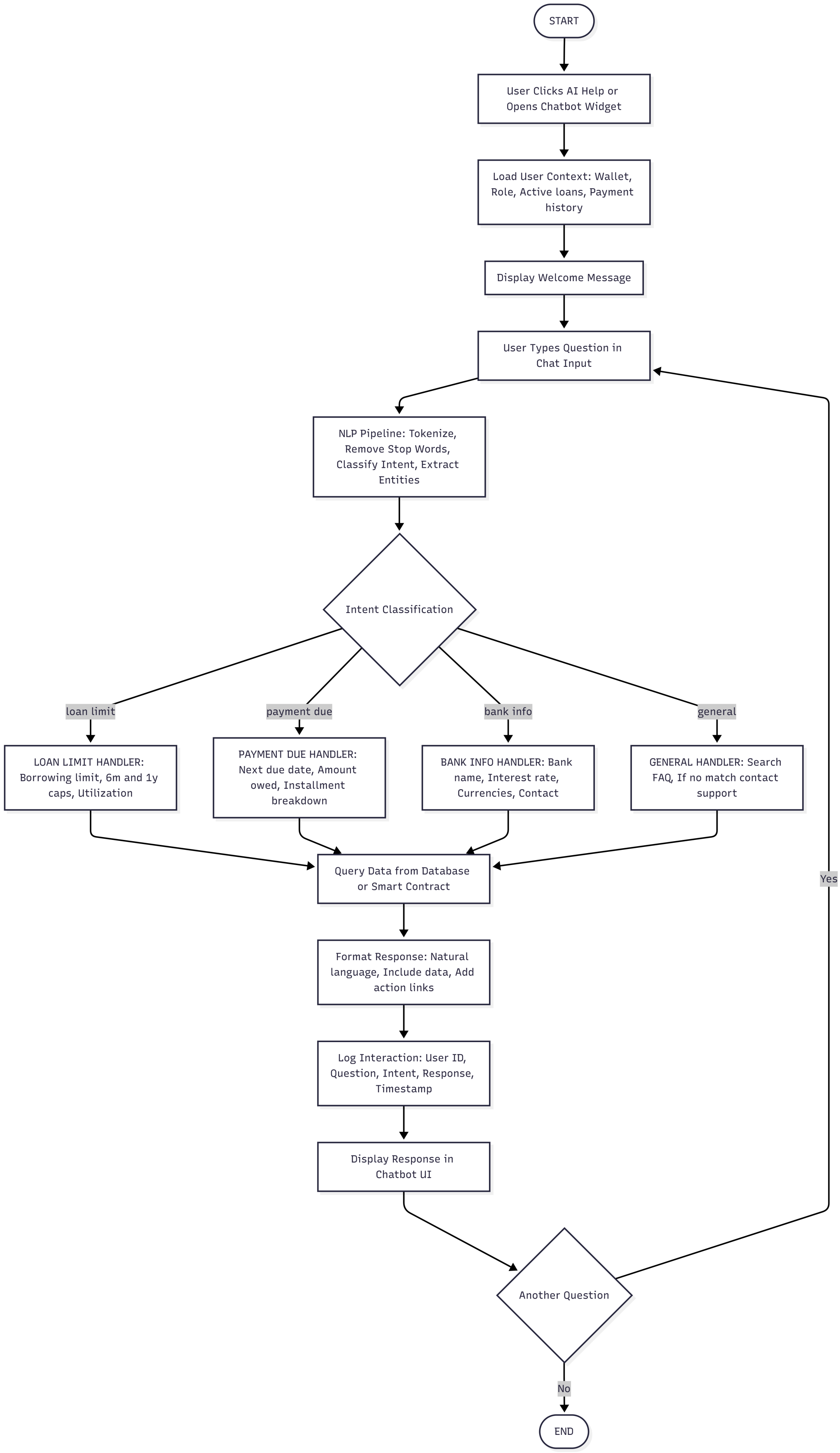

Figure 3.10: Activity Diagram AI Chatbot Interaction Flow

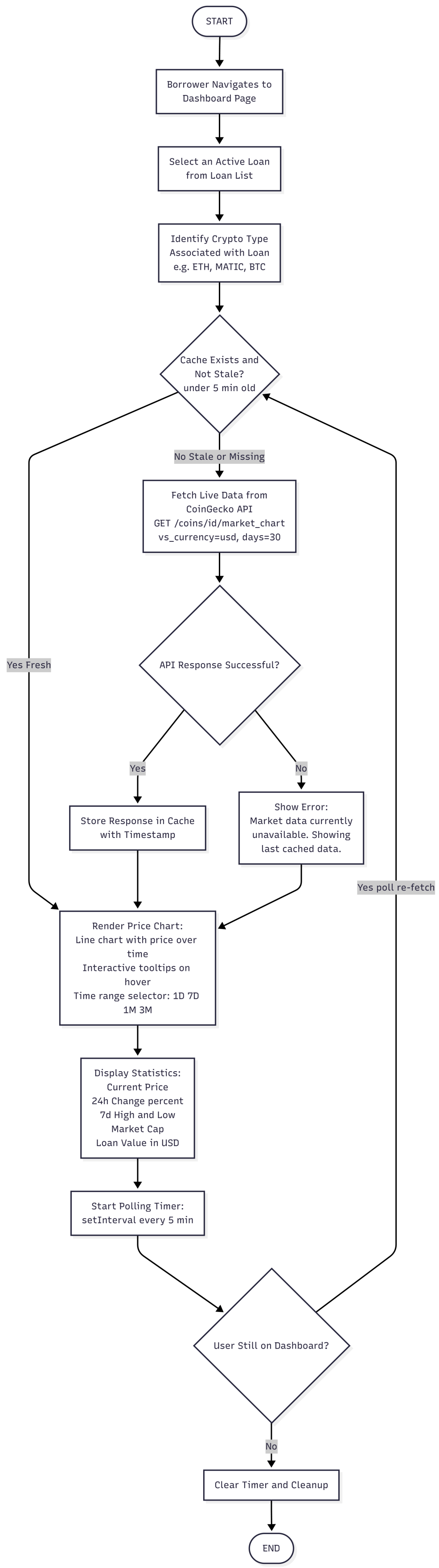

Figure 3.11: activity diagram Market Data Viewing Flow

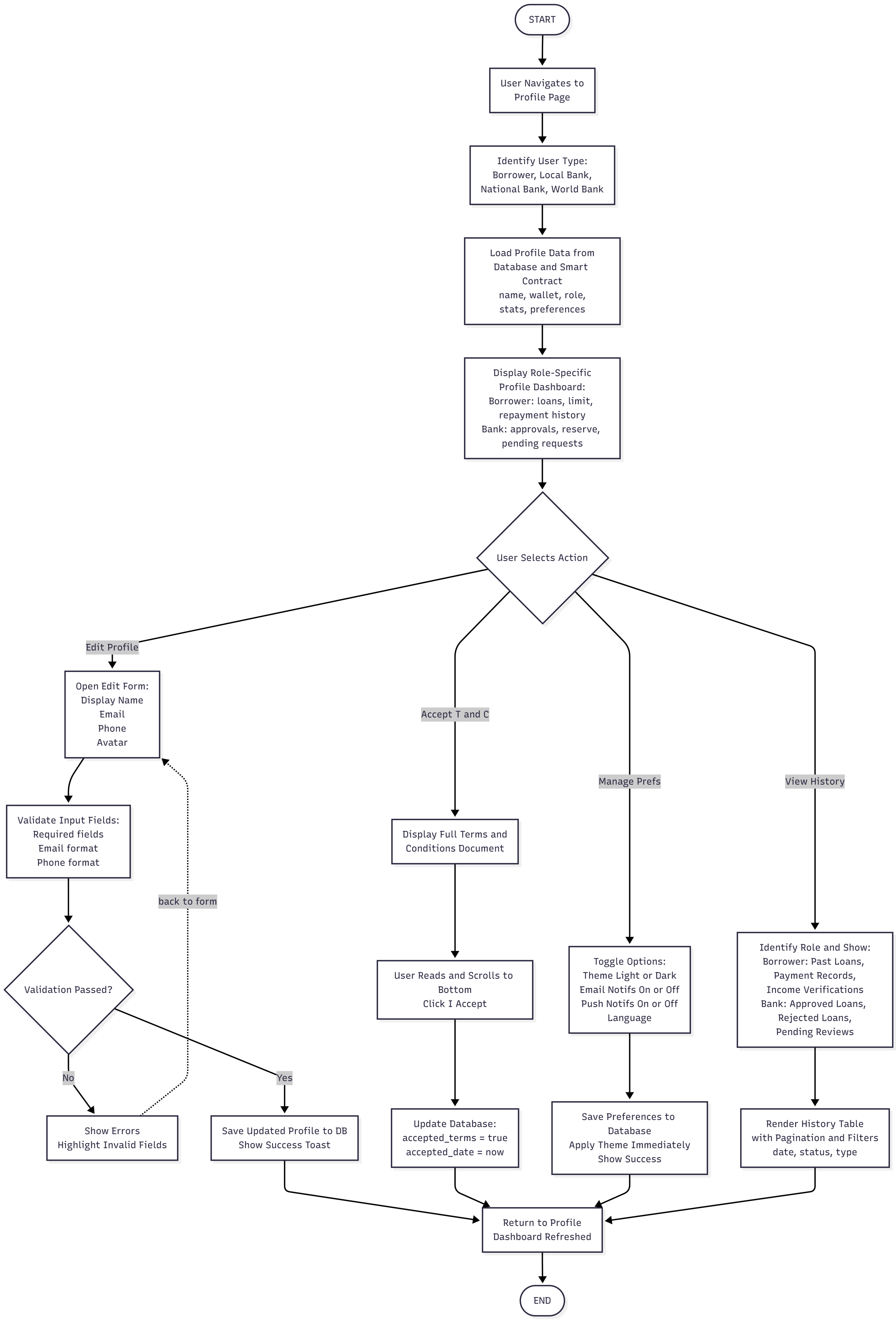

Figure 3.12: Activity Diagram Profile Management Flow

3.6.3 Data Flow Diagrams

Figures 3.13–3.15 present the data flow diagrams at the context level (Level 0) and decomposed level (Level 1).

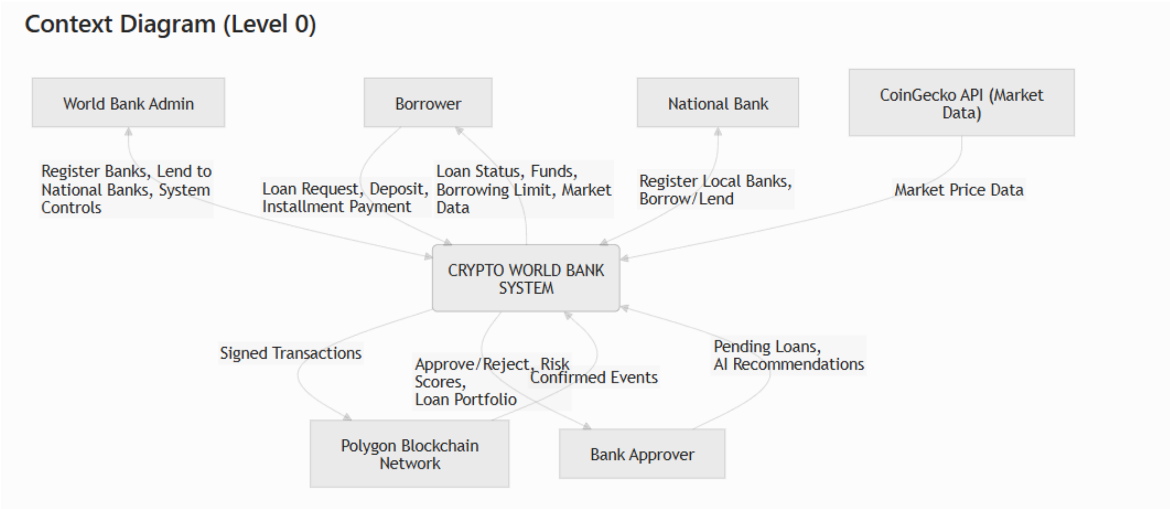

Figure 3.13: Dataflow Diagram (Context Diagram Level - 0)

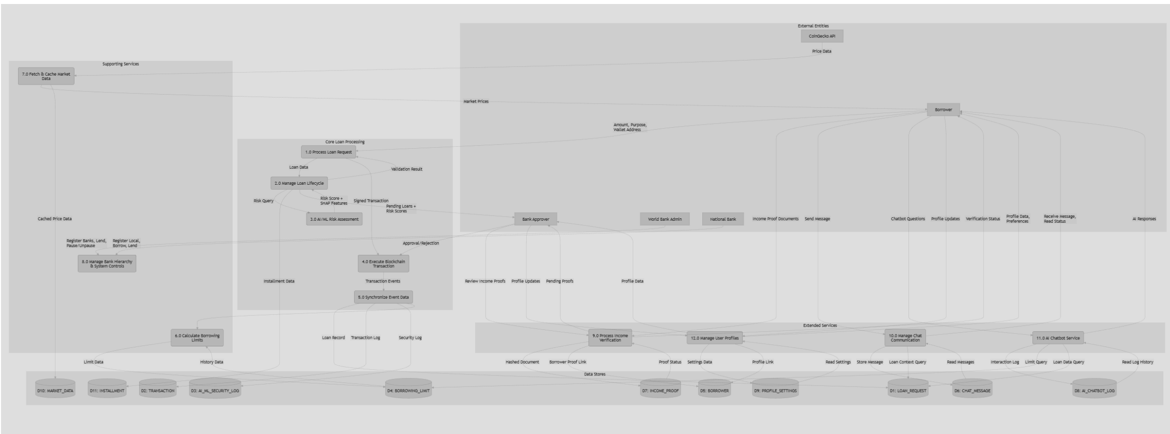

Figure 3.14: Data flow diagram (level - 1)

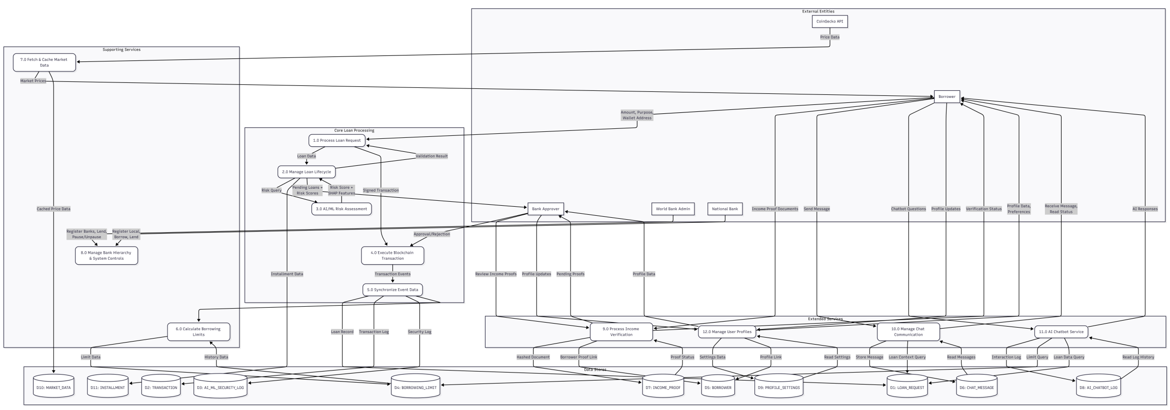

Figure 3.15: dataflow diagram 2 (level -1)

3.6.4 Sequence Diagrams

Figures 3.16–3.24 present the sequence diagrams for the platform's key interaction flows.

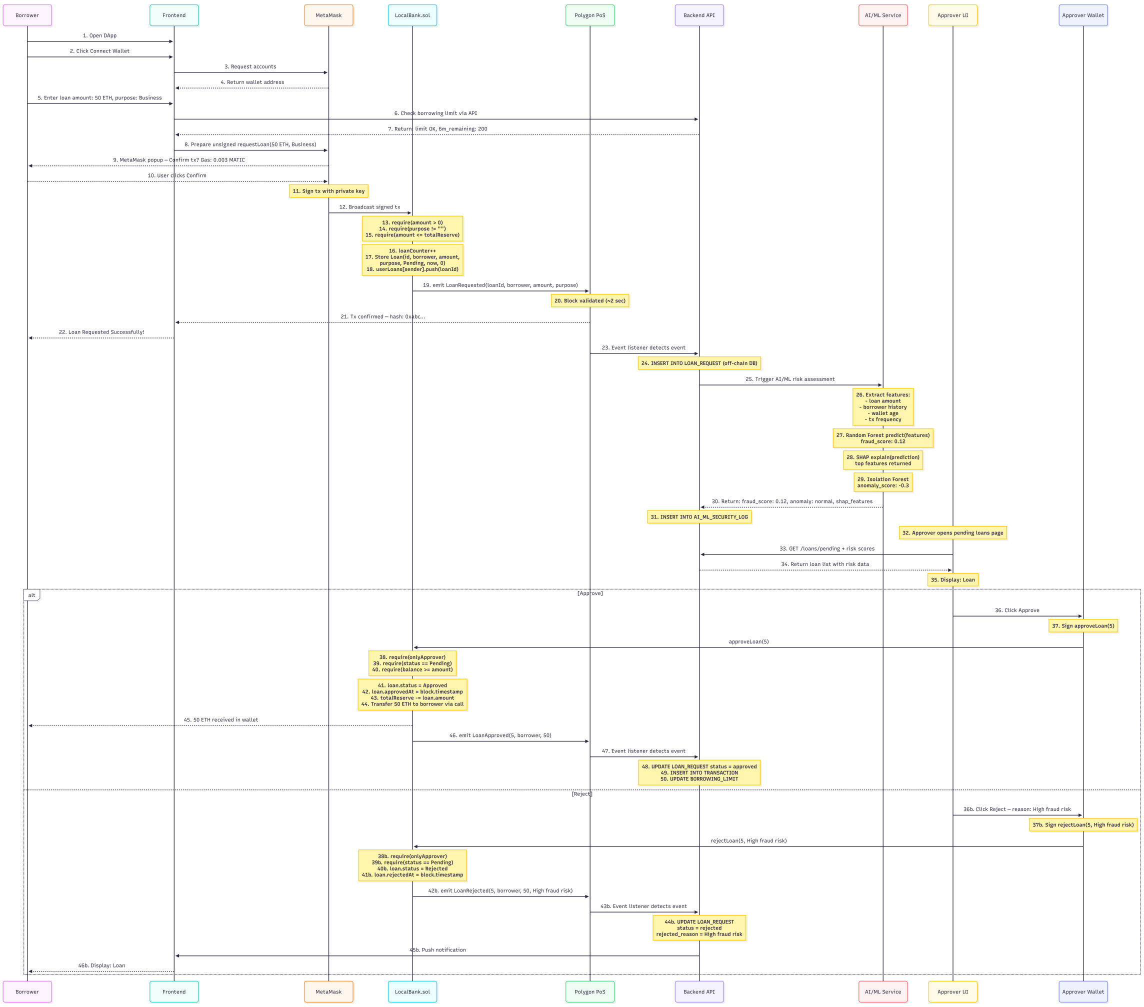

Figure 3.16: Sequence Diagram 1 Loan Request, AI Risk Check, and Approval Decision

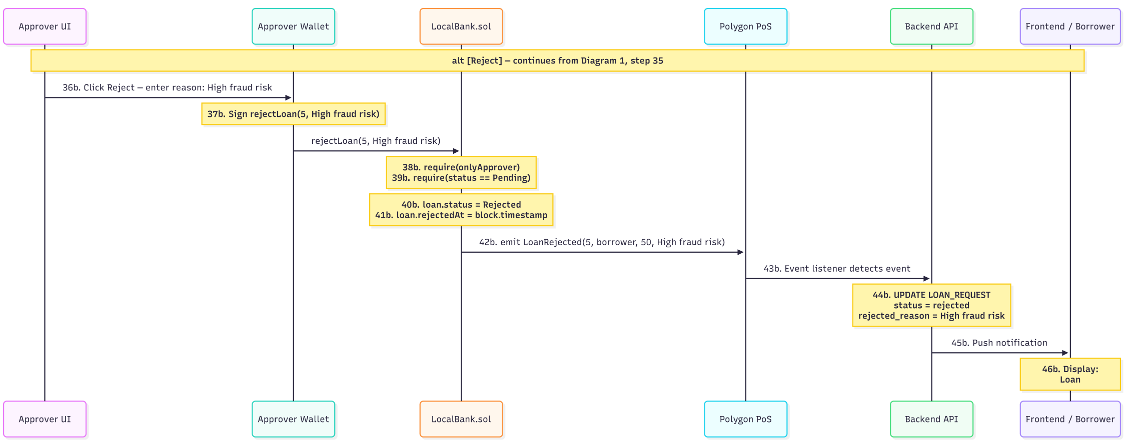

Figure 3.17: Sequence Diagram 1B Reject Path - alt Reject

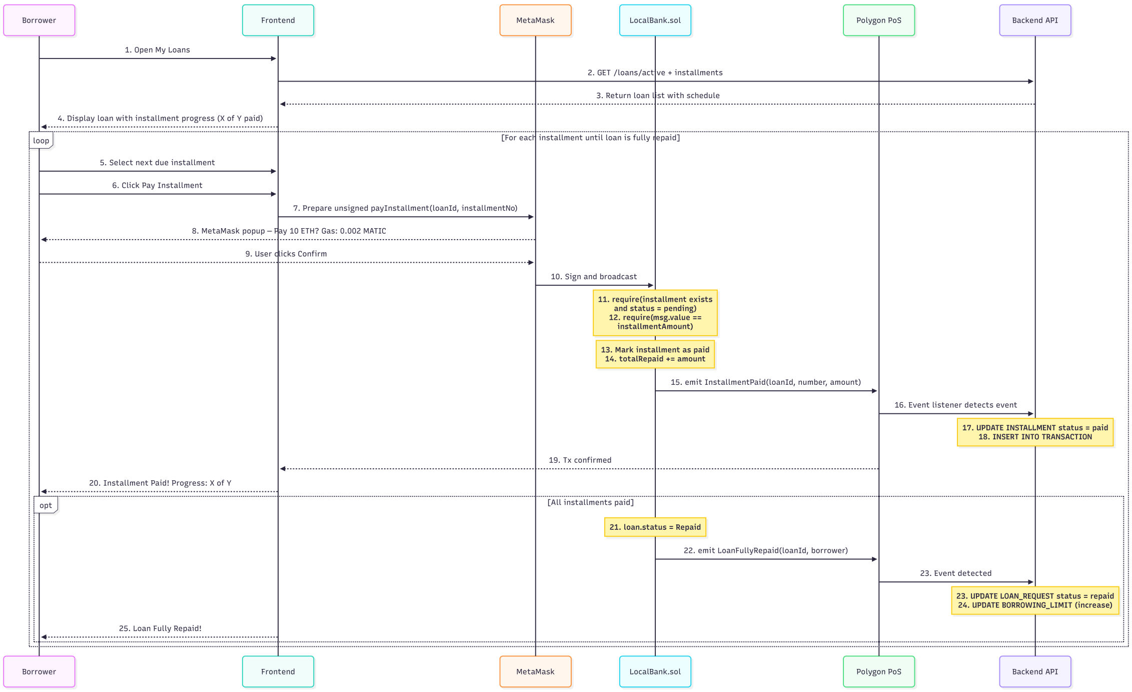

Figure 3.18: Sequence Diagram 2 Installment Payment Loop

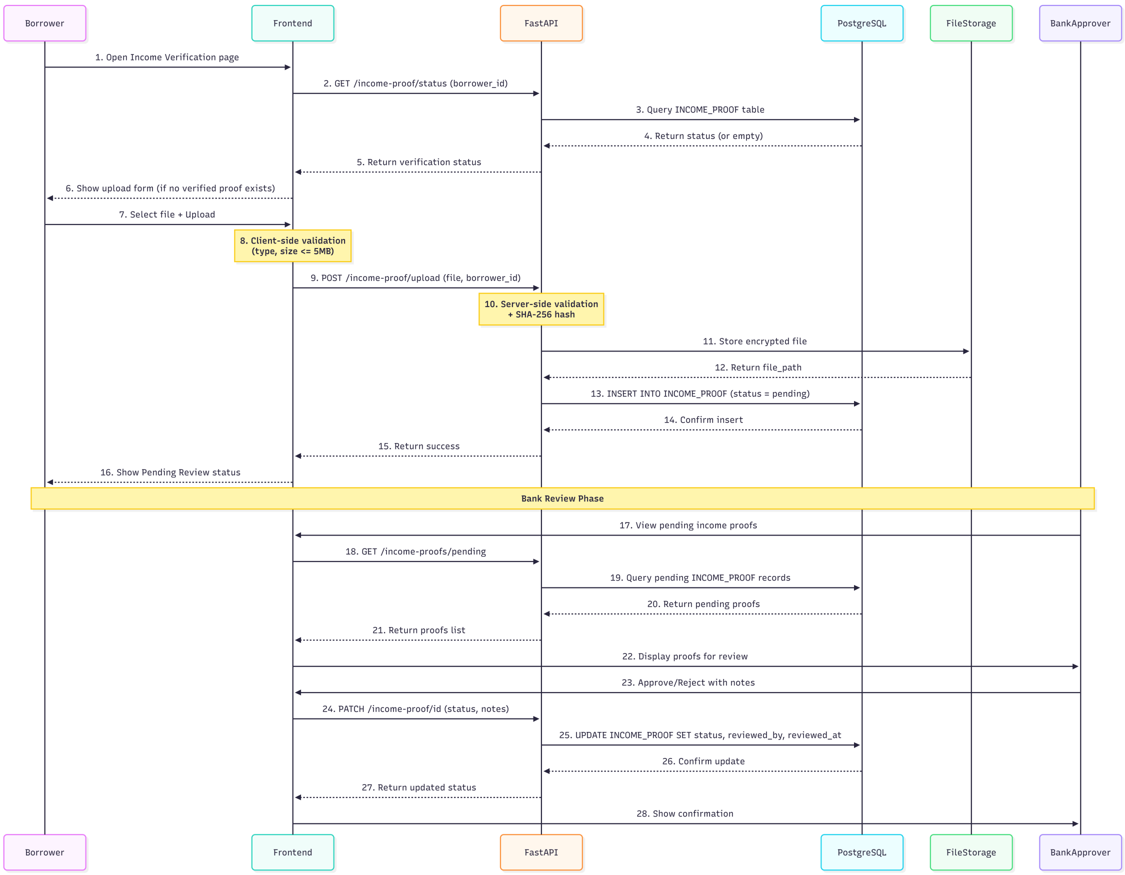

Figure 3.19: Sequence Diagram 3 Income Verification

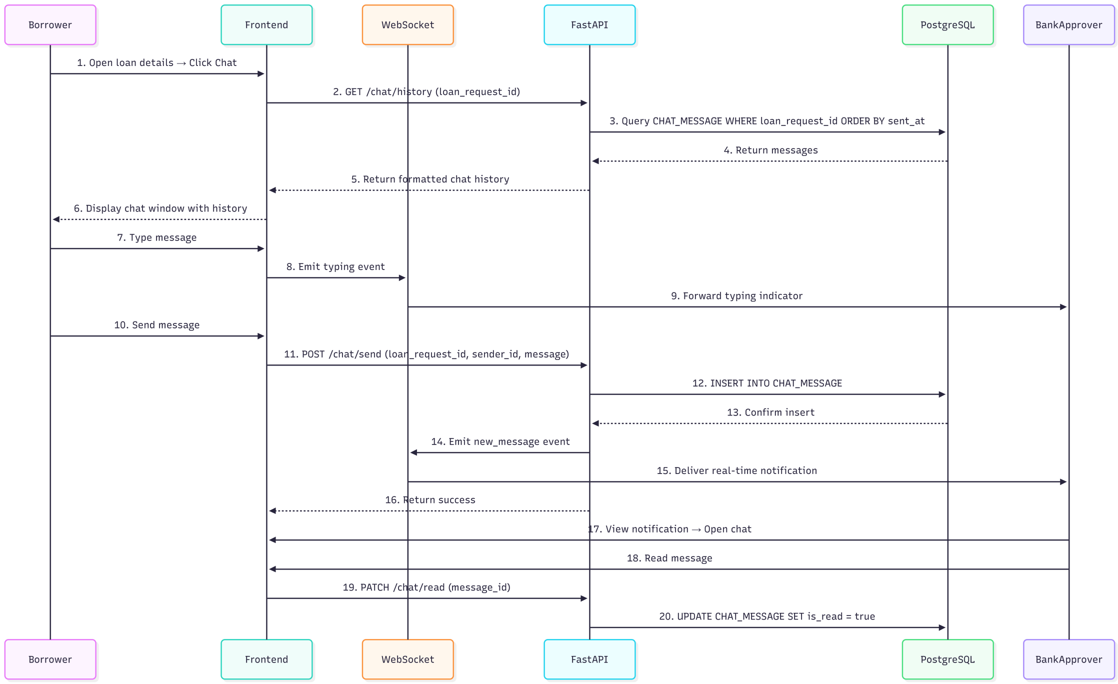

Figure 3.20: Sequence Diagram 4 Chat System

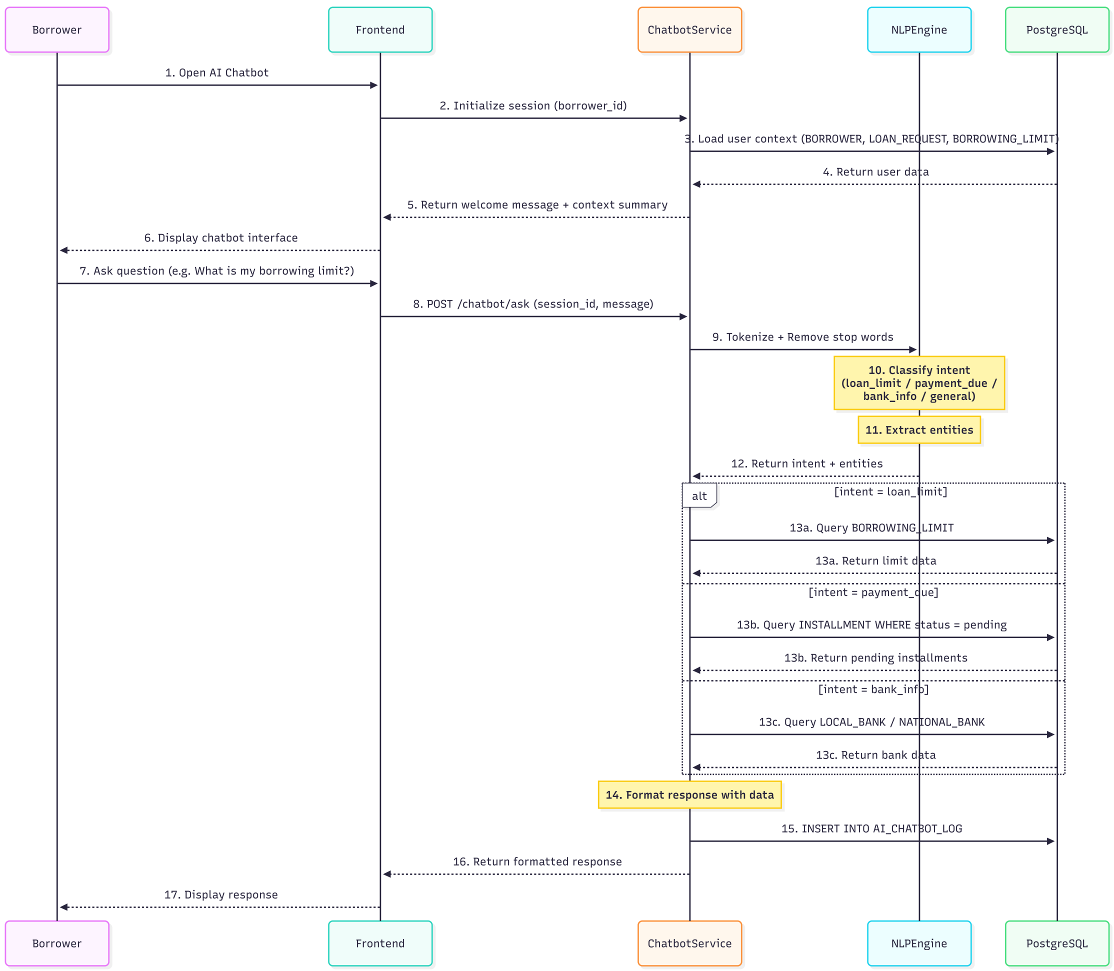

Figure 3.21: Sequence Diagram 5 AI Chatbot Interaction

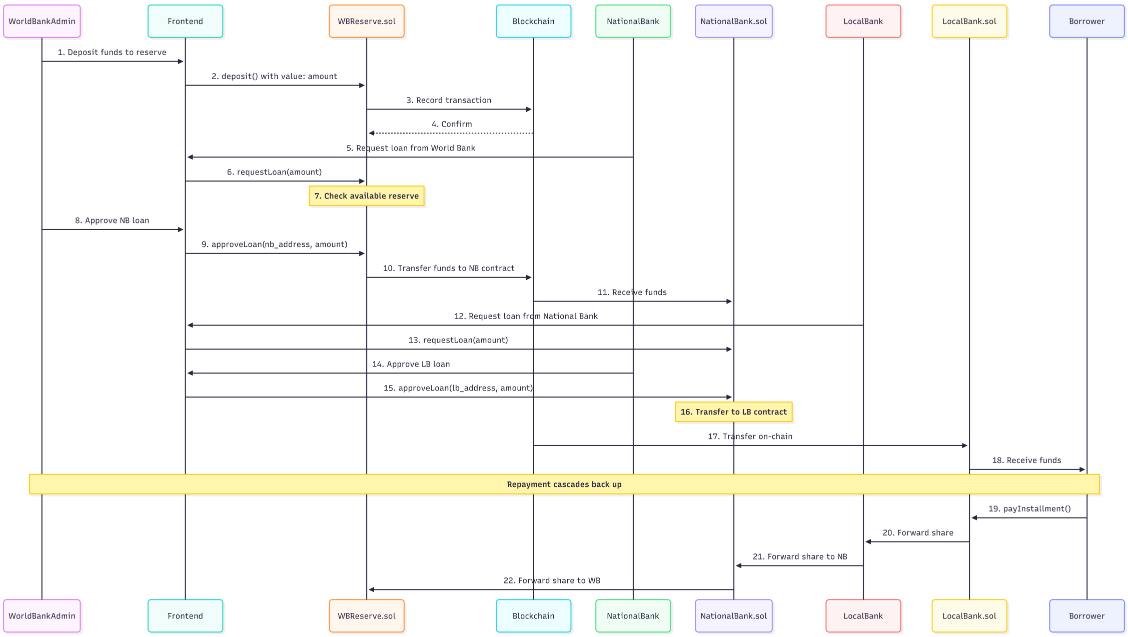

Figure 3.22: Sequence Diagram 6 Hierarchical Banking

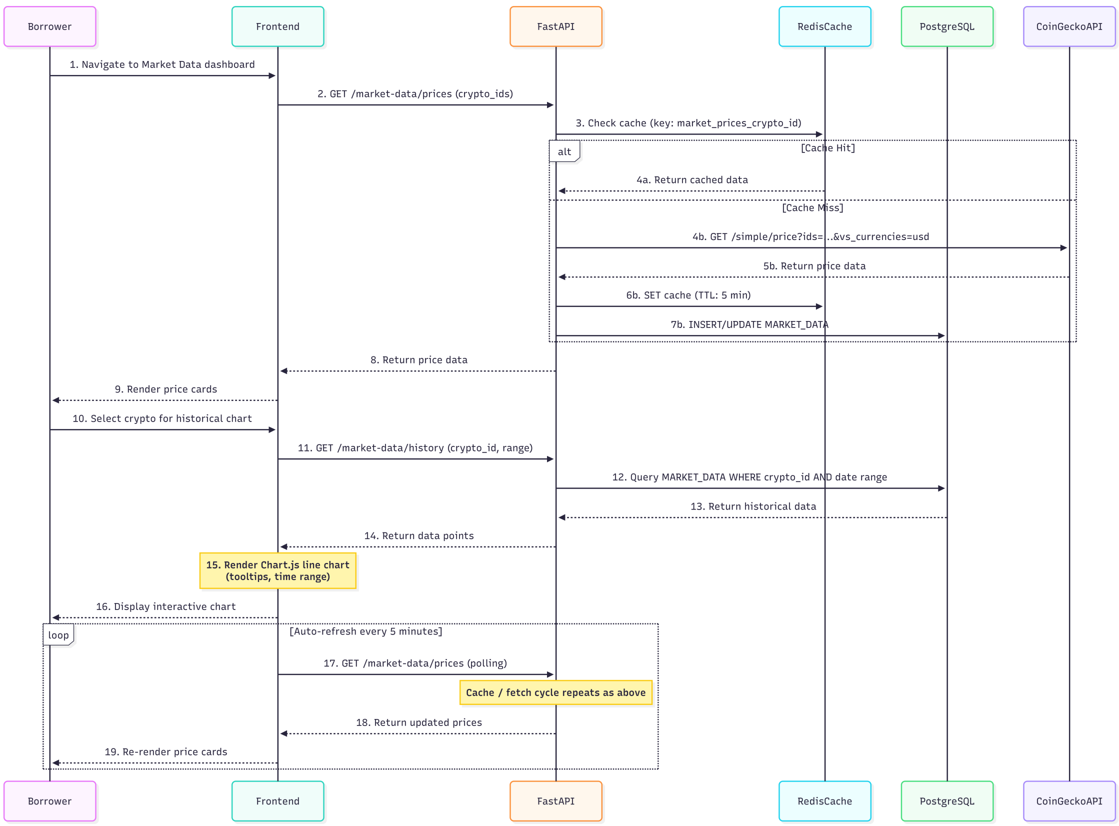

Figure 3.23: Sequence Diagram 7 Market Data Retrieval

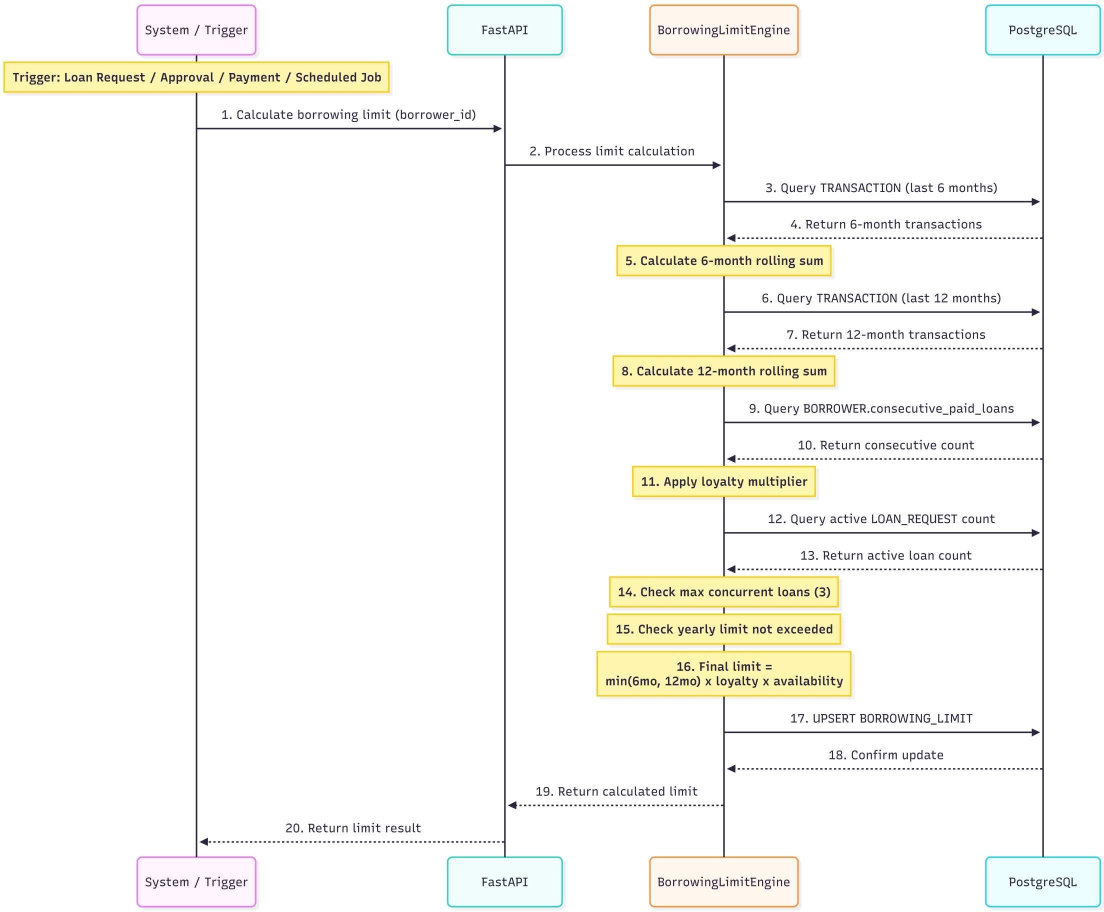

Figure 3.24: Sequence Diagram 8 Borrowing Limit Calculation

3.6.5 Four-Tier Capital Flow

Figure 3.25 illustrates the hierarchical capital flow and cascading repayment structure.

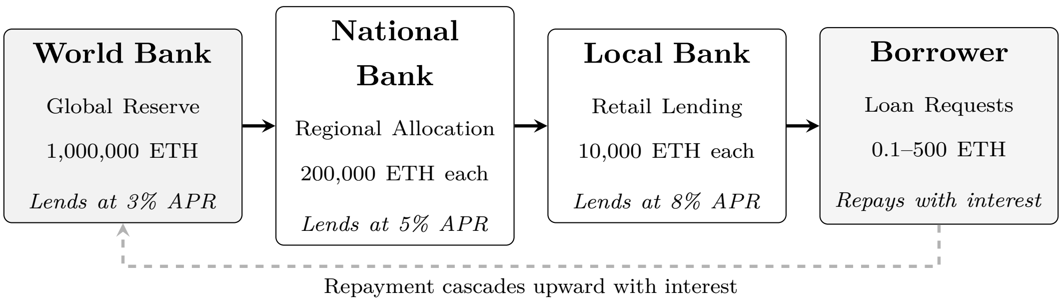

Figure 3.25: Four-tier hierarchical capital flow with cascading repayment.

3.7 Auxiliary Dual-Currency Facility

The cryptocurrency exchange feature of Crypto World Bank is built upon the cryptocurrency exchange facility that is an auxiliary facility incorporated into the current banking infrastructures all over the world. Instead of being a separate exchange, the platform is a service that is provided by all participating banks as an additional service to the existing product portfolio.

- **Integration model:** The dual-currency facility is offered by all participating banks as an add-on service to their existing product portfolio. No additional banking license or separate entity is needed.
- **Eligibility determination:** Eligibility of the dual-currency service is based on the bank officers managing the lending operations, on the basis of existing KYC/AML compliance, account status, and lending relationship history.
- **No defaulting of conditions:** The project does not override, modify, or default any existing banking conditions, regulatory requirements, or contractual obligations.
- **Scope:** The facility enables fiat-to-crypto and crypto-to-fiat conversions, cryptocurrency-denominated lending and repayment, and transparent on-chain transaction records—all within the governance structure of the participating bank.

3.8 Governance Framework

3.8.1 Network Membership Governance

Table 3.8: Network membership governance.

Governance Aspect	Implementation
Member on-boarding	World Bank owner registers National Banks; National Banks register Local Banks; Local Banks designate approvers — all enforced on-chain
Member off-boarding	Deactivation flags in smart contracts; cascading access revocation
Regulatory oversight	Audit log emission via smart contract events; planned read-only regulator dashboard
Permission structure	Hierarchical: Owner → National Bank → Local Bank → Approver → Borrower; enforced by on-chain role-check modifiers
Network operations	Pause/unpause mechanism for emergency response; emergency withdrawal for critical situations

3.8.2 Business Network Governance

Table 3.9: Business network governance.

Governance Aspect	Implementation
Business charter	Defined in project documentation; operational parameters codified in smart contract constants
Common services	Reserve management, loan lifecycle orchestration, event-driven notification system
Business SLA	Testnet phase: best-effort availability. Production phase: 99.5% target uptime with multi-region deployment
Regulatory compliance	Architecture designed for audit trail generation; data partitioning supports GDPR-style data subject requests

3.8.3 Technology Infrastructure Governance

Table 3.10: Technology infrastructure governance.

Governance Aspect	Implementation
Distributed IT structure	Client-side frontend (decentralized delivery); blockchain layer (fully decentralized); backend API (centralized, horizontally scalable)
Technology assessment	Continuous evaluation of EVM alternatives (L2 rollups, sidechains) for cost and performance optimization
On-chain / off-chain data services	Clearly partitioned (see Table 3.5); event listeners synchronize state between layers
Risk mitigation	Smart contract pause mechanism; ReentrancyGuard; input validation; planned formal security audit

3.8.4 Regulatory Compliance Considerations

As the platform is active in the regulated banking sector:

- Our prototype stage would be solely on the **public testnets** without attached real monetary value.
- The architecture would facilitate **audit logs generation** to be reviewed by regulatory regimes on the sandbox programs.
- Future production deployment would engage **regulatory sandbox programs** in target jurisdictions.

3.8.5 Asset Tokenization

- **Current implementation:** The native blockchain currency (ETH/MATIC) is used as the reserve and loan currency.
- **Planned extension:** Support for ERC-20 stablecoins (USDC, USDT) for lending operations in USD; tokenized collateral instruments for lending situations where there isn't enough collateral.

Chapter 4

Methodology

This project and its planning have been structured in accordance with the Blockchain Olympiad Bangladesh AI guidelines [16] and the final-year project evaluation rubrics.

4.1 Development Methodology

We adopted a **lightweight Agile/Scrum** methodology tailored for an academic prototype with a fixed two-month development window. The iterative approach enables incremental delivery of demonstrable features while accommodating evolving requirements inherent in research-oriented development. Figure 4.1 illustrates our Agile process.

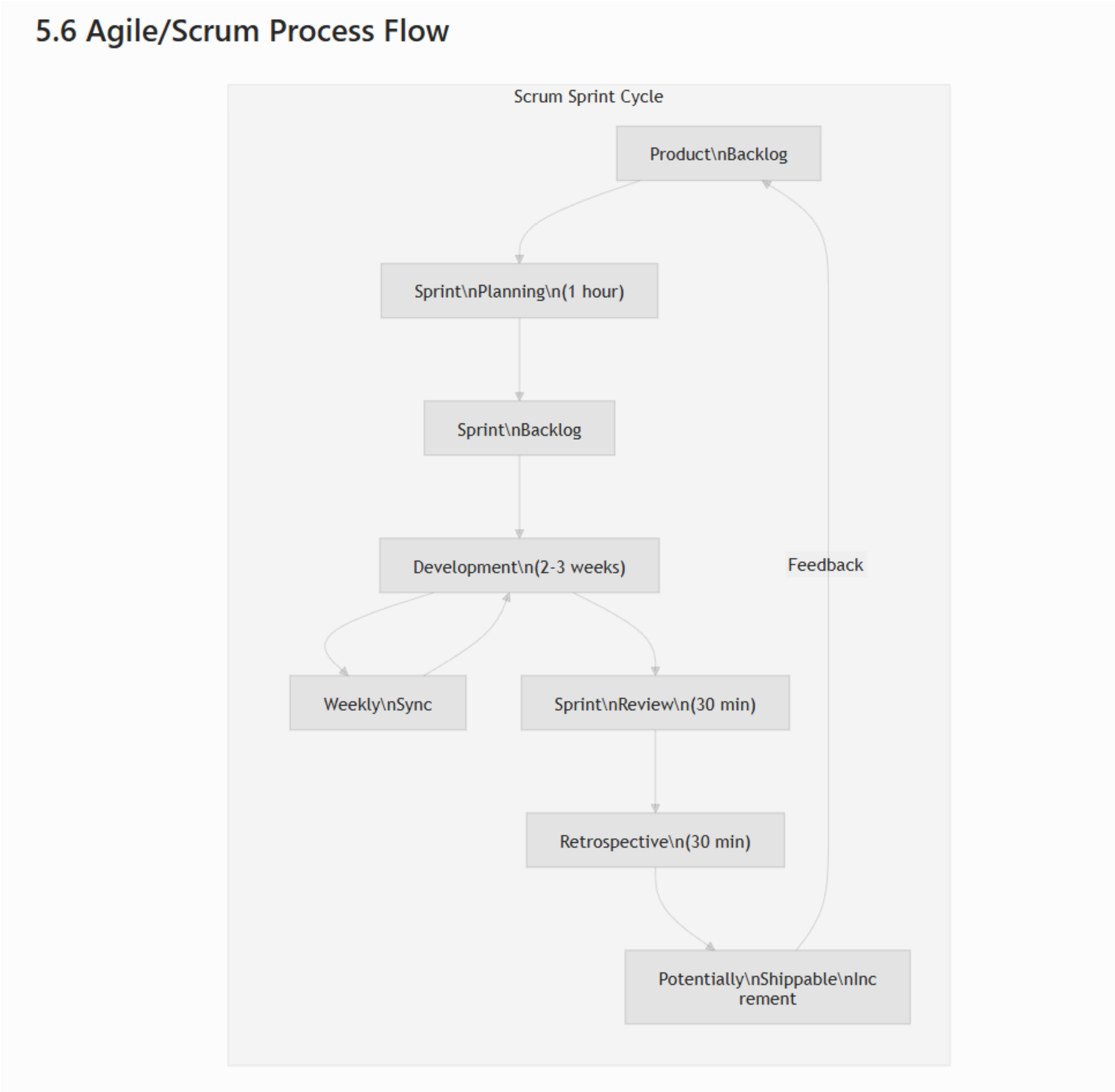

Figure 4.1: Agile/Scrum process diagram showing sprint cycles, backlog refinement, and iterative delivery within the 8-week development window.

Here's how we'll use Agile:

- **Sprint Duration:** A period of 2 to 3 weeks is allocated for every sprint (3 sprints total).
- **Team Size:** A group of two developers will be assigned (thesis project). The work will be focused on the thesis project.
- **Weekly Sync:** A weekly gathering will occur to assess progress. Issues will be identified during these meetings. Plans will also be developed in these sessions alongside progress checks.
- **Sprint Planning:** Planning for the sprint will take place for one hour at the beginning of every sprint.
- **Sprint Review and Retrospective:** Reviewing work occurs at every sprint's

conclusion. A discussion on lessons learned will be conducted for 30 minutes.

4.2 Planned AI/ML Support

The initial section of the thesis involves assistance from AI/ML. Fraud checks, detection of unusual activities and clarification of decisions in lending will be demonstrated. A reliance on an impeccable data system is unwelcome for this paper.

For now AI/ML will be simple:

- **Fraud verification:** Utilizing Random Forest for assessing the risk of transactions. Risk is determined through an analysis by Random Forest.
- **Anomaly detection:** Isolation Forest is recommended for detecting unusual wallet activities when data is limited. It's not necessary to have extensive data for this method to be effective.
- **Explainability:** Utilizing SHAP will continue to provide insights into decision-making. Explanations will be offered for the reasoning behind decisions.

Building comprehensive datasets and preparing models for practical application will be postponed. The focus of the report emphasizes the setup of blockchain lending over developing a complete machine-learning initiative. A complete machine-learning project will not be undertaken at this time.

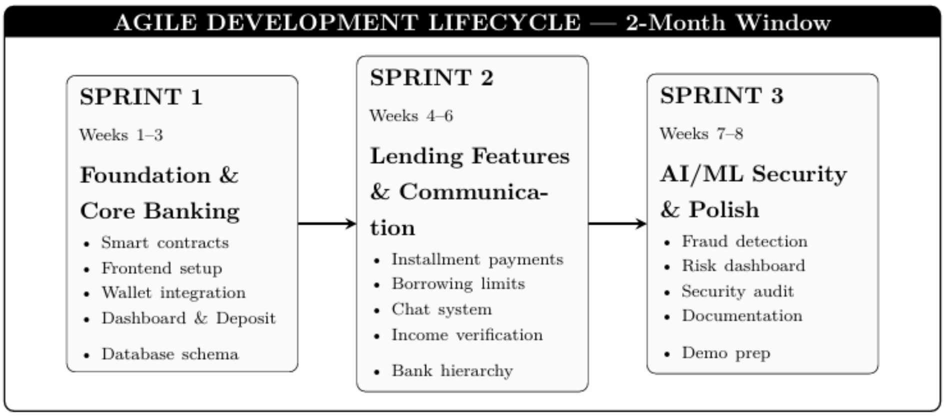

Figure 4.2: Agile development lifecycle — 2-month window with three sprints.

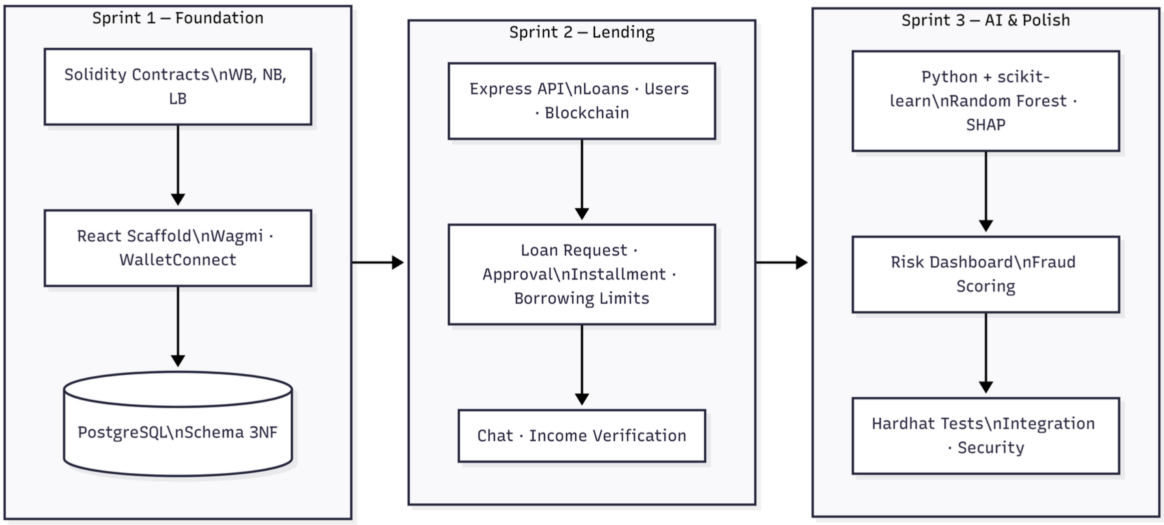

Figure 4.3: Technical development methodology: blockchain (Solidity), frontend (React, Wagmi, WalletConnect), backend (Express.js REST API, PostgreSQL), and AI/ML (Python, FastAPI, scikit-learn, Random Forest, SHAP) components across the three sprints.

4.3 Sprint Plan and Deliverables

4.3.1 Sprint 1: Foundation and Core Banking (Weeks 1–3)

Sprint Goal: Establish core blockchain infrastructure and hierarchical banking structure.

Table 4.1: Sprint 1 backlog: Smart contract development (21 pts).

ID	User Story	Pts
US-1.1	World Bank contract — reserve management, national bank registration	5
US-1.2	National Bank contract — borrow from WB, lend to Local Banks	5
US-1.3	Local Bank contract — borrow from NB, lend to users	5
US-1.4	Role-based access control — WB/NB/LB Admin, Bank User, Borrower	3
US-1.5	Gas cost management — initiator pays; Polygon low-fee ($0.001–0.01/tx)	3

Table 4.2: Sprint 1 backlog: Frontend foundation (13 pts) and database schema (8 pts).

ID	User Story	Pts
US-1.6	Wallet connection — MetaMask, WalletConnect, Sepolia/Amoy	3
US-1.7	Dashboard UI — Material Design 3, responsive, blockchain-themed	5
US-1.8	Navigation and layout — AppBar, role-based menu	3
US-1.9	Blockchain visual elements — tx hash display, security badges	2
US-1.10	Database design — 15 tables, 3NF	5
US-1.11	Migration scripts and seed data	3

Sprint 1 Total: 42 story points. Deliverables: smart contracts deployed on testnet, basic frontend with wallet connection, database schema implemented, role-based access control, dashboard UI.

4.3.2 Sprint 2: Lending Features and Communication (Weeks 4–6)

Sprint Goal: Implement complete lending workflow and communication features.

Table 4.3: Sprint 2 backlog: Lending and communication features (50 pts).

ID	User Story	Pts
US-2.1	Loan request submission with blockchain transaction	5
US-2.2	Loan approval and rejection workflow with bank approver	5
US-2.3	Installment payment system with automated schedule generation	8
US-2.4	Borrowing limit engine (6-month and 1-year rolling windows)	5
US-2.5	Borrower–bank real-time chat system	5
US-2.6	Income verification document upload and review	5
US-2.7	Hierarchical bank registration (National → Local)	5
US-2.8	Bank user management and approver designation	3
US-2.9	Loan history and transaction tracking pages	5
US-2.10	QR code generation for wallet addresses	2
US-2.11	Responsive UI polish and error handling	2

4.3.3 Sprint 3: AI/ML Security and Finalization (Weeks 7–8)

Sprint Goal: Integrate AI/ML analytics, complete testing, and prepare documentation.

Table 4.4: Sprint 3 backlog: AI/ML security and finalization (38 pts).

ID	User Story	Pts
US-3.1	Random Forest fraud detection model training and deployment	8
US-3.2	SHAP-based explainability for risk assessments	5
US-3.3	Isolation Forest anomaly detection for wallet behavior	5
US-3.4	Risk dashboard with real-time AI/ML scores	5
US-3.5	AI chatbot for borrower assistance	3
US-3.6	Market data visualization page	3
US-3.7	Profile and settings management	2
US-3.8	Security audit and vulnerability assessment	3
US-3.9	Documentation and report finalization	2
US-3.10	Demo preparation and presentation	2

Figure 4.4 shows the sprint submission workflow.

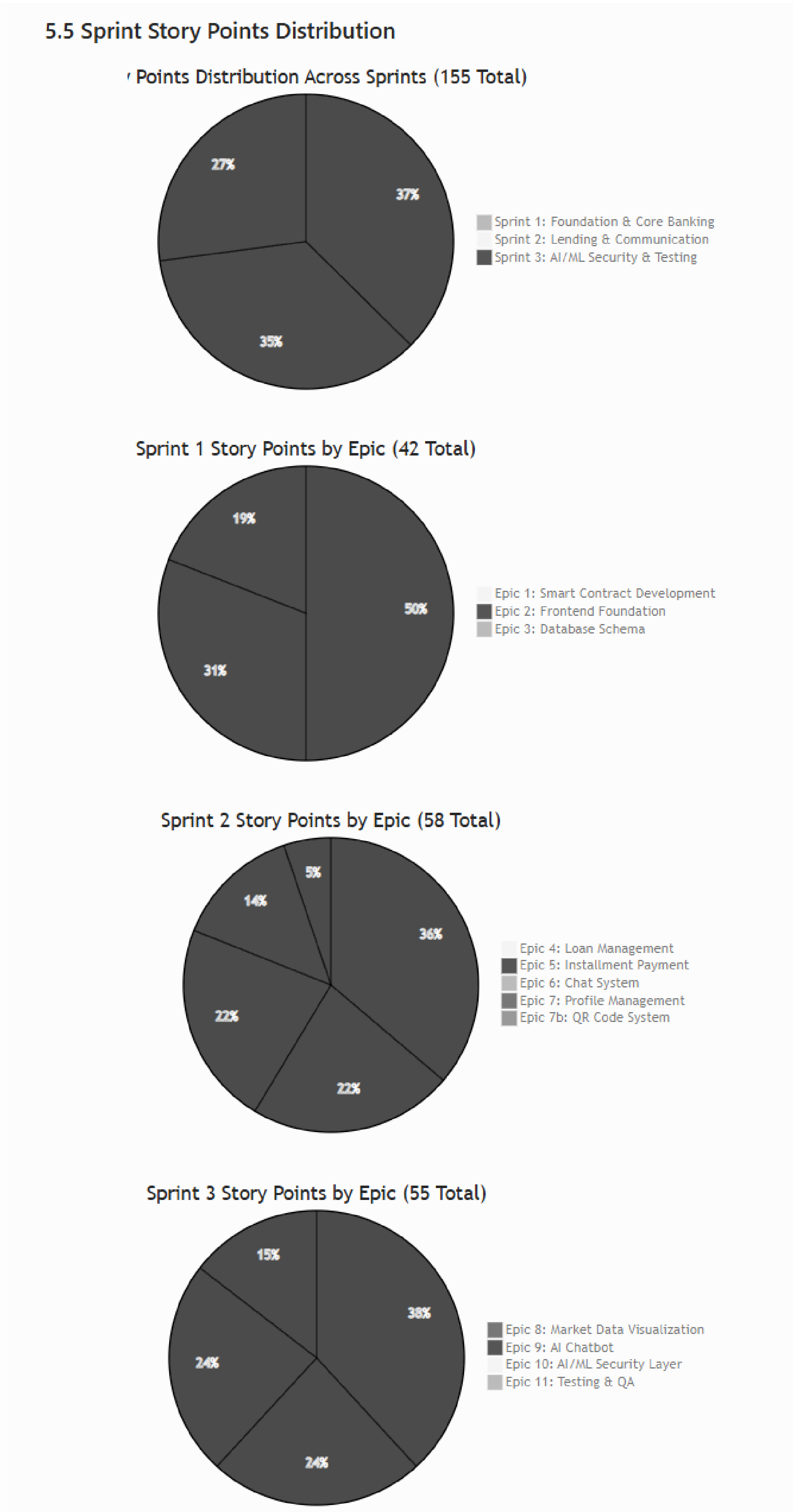

Figure 4.4: Sprint submission diagram showing the workflow from backlog refinement through development, testing, review, and delivery for each sprint cycle.

4.4 SDLC Stage Mapping

We map our Agile sprints to the seven stages of the Software Development Life Cycle (SDLC). Figure 4.5 illustrates this mapping.

Table 4.5: SDLC stage mapping.

SDLC Stage	Project Activity	Deliverable
1. Planning	Feasibility studies; professor consultations; guideline review	Feasibility Report; Project Plan
2. Requirements	System analysis; use case definitions; constraints identification	Use case diagrams; UC-1 through UC-5
3. Design	Three-layer architecture; DB schema (3NF); smart contract interfaces	Architecture diagrams; ERD; DFD
4. Development	Sprint 1–3: smart contracts, frontend, backend, AI/ML	Source code; DApp prototype
5. Testing	Hardhat unit tests (12+); integration testing; AI/ML evaluation	Test reports; model metrics
6. Deployment	Testnet deployment; frontend (Vercel); backend (Render)	Live prototype on testnet
7. Maintenance	Monitoring; model retraining; bug fixes; iteration	Updated docs; retrained models

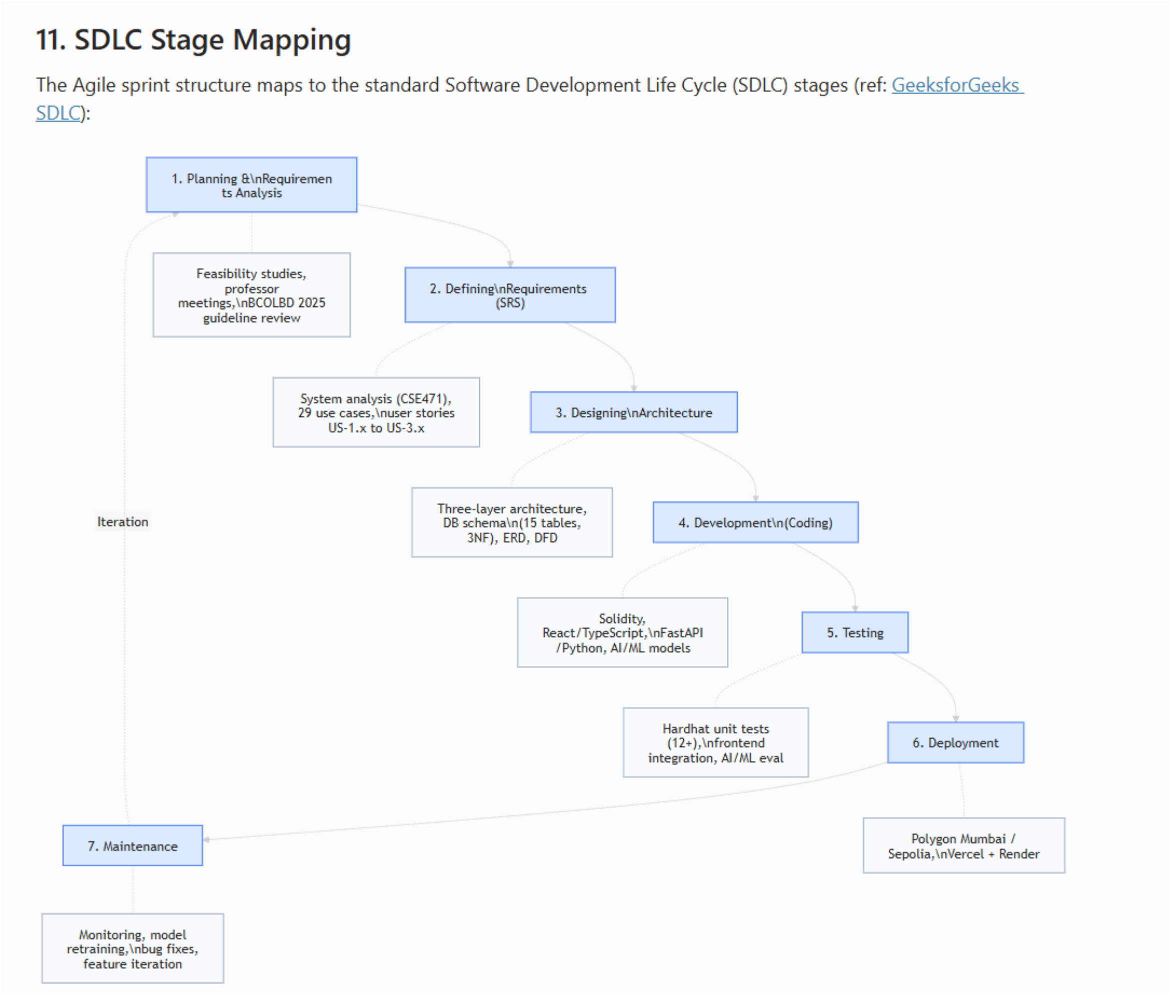

Figure 4.5: SDLC stage mapping diagram showing alignment between the seven SDLC stages and our three-sprint Agile methodology.

4.5 Design Decisions and Alternatives

For each major design decision, we evaluated alternatives and justified selections based on technical criteria, ecosystem maturity, and project constraints. Figure 4.6 visualizes these comparisons.

Table 4.6: Design decisions and alternatives considered.

Decision Area	1st Choice	2nd Choice	Key Criterion
Methodology	Agile / Scrum	Incremental	Evolving scope, milestones
Architecture	DApp + Off-chain AI	Hybrid w/ Oracle	Gas cost, ML flexibility
Frontend	React + TypeScript	Vue + TypeScript	Web3 ecosystem maturity
Smart Contract	EVM (Solidity)	Solana (Rust)	Ecosystem size, free testnets
UI Design	MUI (Material 3)	Tailwind CSS	Banking UI conventions
Fraud Detection	Random Forest	XGBoost	SHAP compatibility, implementation simplicity
Anomaly Detection	Isolation Forest	Autoencoder	Unsupervised, no labeled data needed
XAI Method	SHAP	LIME	Theoretical guarantees, regulatory fit
Database	PostgreSQL	SQLite	3NF support, async queries
Hosting	Vercel + Render	Localhost only	$0 cost, publicly accessible URL

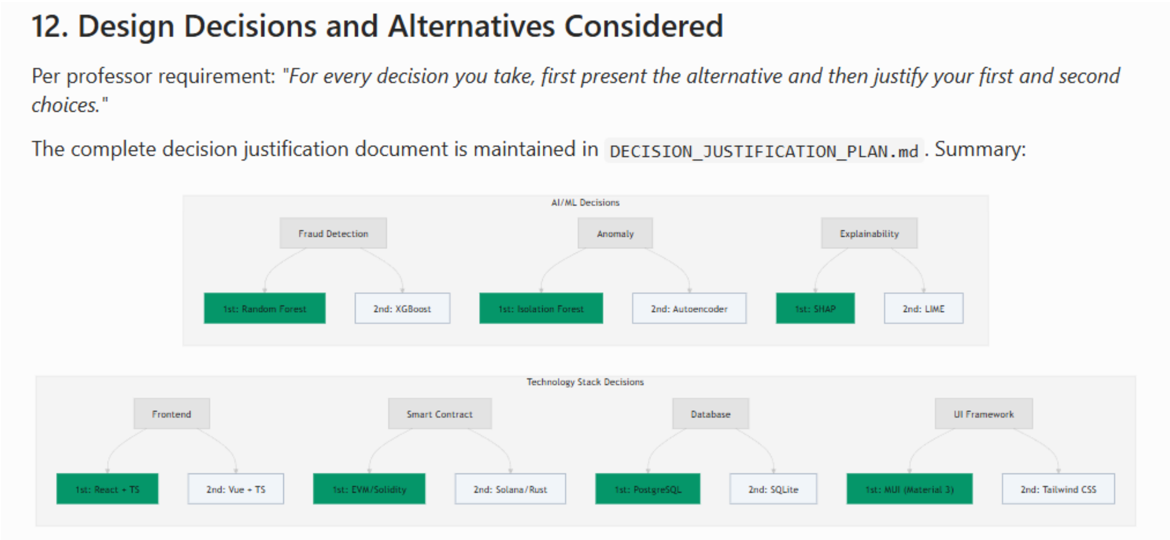

Figure 4.6: Design decisions and alternatives visualization across all major technology choices.

4.5.1 Justification of Selected Technologies

1st Choice Justifications:

- **Agile/Scrum** was necessary due to the project's evolving nature. Frequent updates to smart contract designs, user interface steps and AI/ML connections were required. Adopting Agile allows for plan alterations during brief work cycles effectively avoiding time and resource loss. Significance arises as new concepts emerge during the prototype development process.
- **DApp + Off-chain AI** architecture allows intensive machine learning operations to occur outside the blockchain. Machine learning tasks are often run on the blockchain at a considerable expense occasionally exceeding $100 for a single operation. Off-chain processing enables the use of rapid GPUs along with widely-used Python machine learning resources. Final lending choices are managed exclusively by smart contracts on-chain ensuring that trust is maintained in critical areas.
- **React + TypeScript** is beneficial as many Web3 libraries such as Wagmi, RainbowKit, Viem, and ethers.js are supported seamlessly. A vast community of developers exceeding 10 million exists for React making it simpler to access assistance and solutions.
- **EVM (Solidity)** is regarded as the most thoroughly examined smart contract system. Over $100 billion is secured across various blockchains by it. Building and testing the project without incurring costs is possible with free test networks such as Sepolia and Amoy. Security tools provided by OpenZeppelin have been thoroughly tested to ensure the protection of contracts.
- **MUI (Material Design 3)** provides ready-made components for typical banking

interface designs. The package contains various items such as data tables, form checks and layouts for dashboards. Utilizing MUI results in a time saving of around 40% compared to starting from scratch.

- **Random Forest** performs effectively alongside SHAP's TreeExplainer which swiftly computes precise Shapley values for tree models. Precise calculations are essential for adhering to regulations concerning decision-making transparency. Avoiding approximate methods ensures explanations remain stable which prevents issues during audits.
- **Isolation Forest** does not require labeled data for training. Labeled fraud samples are few making it suitable for DeFi lending. Anomalies in transactions are scored swiftly as it operates with a time complexity of $O(n \log n)$.
- **SHAP** provides feature importance through solid mathematical foundations from game theory (Shapley values). Characteristics such as accuracy, management of absent data and consistency are not assured by tools like LIME. These properties assist in fulfilling emerging regulations for explainable decisions within financial services.
- **PostgreSQL** effectively manages complex time-based queries. Borrowing limits can be checked over periods such as 6 months or 1 year. ACID compliance is fully supported to ensure financial data remains secure. It supports JSON which facilitates easy adjustments to the database structure during the prototyping phase.
- **Vercel + Render** facilitates deploying the demo globally at no charge. A live prototype can be accessed without the necessity of a setup on personal computers. Supervisors, examiners and other individuals are not required to deal with installation issues.

2nd Choice Justifications (Why Retained as Alternatives):

- **Incremental methodology** enables more predictable outcomes. Results from this method are often delivered with clarity and reliability. Adapting to changes arising from research is not a strength of this approach. Effectiveness occurs mainly in stable environments such as during the production phase.
- **Hybrid with Oracle** architecture (e.g., Chainlink) enables machine learning predictions on-chain by utilizing oracle data. Increased trust is provided for AI decisions. However some delays (30–60 seconds per update) and additional costs ($0.10–1.00 per call) are associated with the use of oracles. Availability hinges on the existence of third-party oracle services.
- **Vue + TypeScript** is relatively straightforward to learn. Its Web3 tools such as vue-dapp are often considered less developed. Updates on wallet integration libraries are not abundant.

- **Solana (Rust)** possesses the capability to process significantly more transactions each second (65,000 TPS) compared to Ethereum's 15 TPS. A smaller developer community is associated with it and the tools available are not as refined. Learning Rust is not easy for everyone. Completing the project in under 8 weeks is not achievable.
- **Tailwind CSS** offers enhanced design flexibility through utility classes. Creating consistent banking-style UI patterns requires significant effort compared to the immediate solutions offered by MUI.
- **XGBoost** frequently outperforms Random Forest in structured data evaluations. Approximate methods are utilized by its SHAP tool rather than precise tree computations. Explanations cannot be guaranteed to be consistent.
- **Autoencoder** can capture complex non-linear anomaly patterns. A large volume of training data and precise tuning are required for its effectiveness. In contrast parameters are not necessary for Isolation Forest. It often suits simple prototypes that work with limited data.
- **LIME** offers faster results. Explanations however lack stability. Different outcomes can arise from identical inputs on multiple checks as demonstrated by Adom et al. [4].
- **SQLite** is straightforward to set up. Multiple writes at one time are not supported. Advanced queries such as window functions and CTEs are not included for calculating borrowing limits.
- **Localhost only** eliminates hosting dependencies. Sharing the project remotely or collaborating for testing is limited by this approach. Academic review won't benefit from such isolation.

4.6 Design Patterns

Our system uses three design patterns:

- **Singleton:** A singleton means that each primary contract is established a single time. Only one version is utilized by all ensuring a unified source of truth. Not everyone can create multiple versions of the contract.
- **Observer:** Contracts generate alerts whenever actions occur such as loan requests. These alerts are processed by a service that updates the database. Notifications are not ignored; they receive attention.
- **Adapter:** An Adapter is utilized by the wallet component to function with various wallet types (like MetaMask). Different wallets that adhere to the EIP-1193 standards can be easily integrated.

4.7 Software Testing Strategy

We test in a few ways:

- **Smart contract unit tests:** Numerous tests for smart contracts exist in our Hardhat environment. Over 12 tests are conducted to verify actions related to managing reserves, sign-ups, loans and access permissions. No scenarios such as depositing zero or executing disallowed actions are missed in these assessments.
- **Integration testing:** Whole process checks are conducted for integration tests. Wallet connection, loan request, approval and repayment processes are evaluated on the network. No critical steps are excluded from the process.
- **AI/ML module verification:** Ensuring functionality is vital for our fraud and unusual activity detectors. Results are generated by the integration of the detectors with the overall system.
- **Frontend testing:** Website appearance is evaluated on Chrome, Firefox and mobile devices. Tests are conducted to ensure that functionality is maintained across various platforms.

Chapter 5

Market Analysis and Feasibility

5.1 Market Sizing

Table 5.1: Market segments with supporting data.

Segment	Description	Estimated Scale
Total Addressable Market (TAM)	Global DeFi lending ($55B+ TVL [13]); cross-border remittances ($860B annual [26]); SME financing gap ($4.5T [20])	$55B–5T+
Serviceable Addressable Market (SAM)	Institutional and semi-institutional lending requiring hierarchical structures; emerging-market credit demand	$5–15B
Serviceable Obtainable Market (SOM)	Pilot deployments in regulatory sandboxes; academic prototypes; NGO-backed microfinance programs	$50–200M

Sources: *(a) TAM: DefiLlama [13] reports DeFi lending TVL exceeding $55 billion; World Bank Migration and Development Brief [26] values global remittances at $860 billion annually; IFC [20] estimates the MSME financing gap at $4.5 trillion; (b) SAM and SOM: project-derived estimates based on industry segmentation of institutional lending and regulatory sandbox addressable market [18].*

Currently no organized lending system in DeFi exists. Simple shared pools are commonly utilized within DeFi. A functioning cross-border money transfer mechanism is not being provided by Ripple ($847M daily) [25] or JPMorgan Kinexys ($3–7B daily) [40]. Large organizations demonstrate a desire for blockchain to be utilized in financial matters. A lending system is being developed by the Crypto World Bank to create accessibility similar to DeFi while also incorporating traditional banking structures aimed at fostering growth. No existing solution completely meets this need.

5.2 Target Customer Segment

The Crypto World Bank targets the **retail customer segment**—individual borrowers and small businesses seeking transparent, accessible crypto-based lending services.

Table 5.2: Target customer segment profile.

Characteristic	Description
Primary Users	Individual retail borrowers seeking personal or small business loans
Geographic Focus	Developing economies with limited traditional banking access (e.g., Bangladesh, Southeast Asia, Sub-Saharan Africa)
Loan Size Range	Micro to mid-range: 0.1 ETH – 500 ETH equivalent (~$200 – $1,000,000 at current rates)
User Profile	Digitally literate individuals with cryptocurrency wallet access; small business owners; gig-economy freelancers
Key Pain Points	High interest rates from informal lenders; lack of credit history; exclusion from banking due to documentation barriers

5.3 Partner Ecosystem

Table 5.3: Partner categories and roles.

Partner Category	Functional Role	Blockchain-Mediated Incentive
Financial Regulators	Regulatory sandbox approval; compliance oversight	Reduced enforcement cost through on-chain transparency
Banking Institutions	Network membership as National/Local Banks	Access to diversified global reserve; reduced settlement friction
Payment Gateways	Fiat-to-crypto on-ramp and off-ramp services	Volume-based transaction fees; expanded market reach
Academic Institutions	Validation of AI/ML models; research publication	Access to anonymized datasets; collaborative opportunities
NGOs	Pilot deployment with underserved populations	Transparent, low-friction credit access for beneficiaries

5.4 Competitive Landscape

The Crypto World Bank operates at the intersection of four distinct competitor categories. Table 5.4 provides a detailed comparison of representative projects in each category, with current metrics as of March 2026.

Table 5.4: Detailed competitive landscape analysis.

Project	**Category**	**Scale (2026)**	**Architecture**	**Gap We Address**
Aave v3 [27]	DeFi lending	$26.3B TVL; 10+ chains	Flat pool; overcollateral-ized; single-tier	No hierarchy; no tiered borrower access; no AI risk layer

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Table 5.4: Detailed competitive landscape analysis (continued).

Project	Category	Scale (2026)	Architecture	Gap We Address
Compound v3 [29]	DeFi lending	$1.4B TVL	Single-borrowable-asset per market	No hierarchy; declining market share; no institutional features
MakerDAO / Sky [30]	Stablecoin / CDP	$6B TVL; $611M rev. target	CDP model; not peer-to-peer lending	Creates money, not a lending system; governance concentration risks
Morpho [43]	DeFi lending	$6.8B TVL; 1.4M users	Isolated markets; peer-to-peer rate matching	Flat primitive; no cross-market hierarchy; no banking integration
Maple Finance [31]	Institutional credit	$2.6–3.8B TVL	Pool Delegate model; under-collateralized	Single-tier; crypto-native borrowers only; no retail access
Goldfinch [32]	Emerging-market credit	$680M originated; 18+ countries	Trust-through-consensus; senior/junior tranches	B2B only (lends to fintechs, not end users); no interbank lending
Ripple / RLUSD [25]	Banking rails	$847M/day cross-border	Payment rail; no lending capability	Moves money between banks but has no lending, deposits, or credit system

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Table 5.4: Detailed competitive landscape analysis (continued).

Project	Category	Scale (2026)	Architecture	Gap We Address
JPMorgan Kinexys [40]	Banking rails	\$3–7B daily volume	Permissioned; single-bank control	Centralized; proprietary; restricted to JPMorgan clients
Stellar [44]	Financial inclusion	\$55.6B annual payment volume	Open payment network; anchors for fiat	Payment network only; no lending, reserves, or interest rate markets
Celo / Mini-Pay [42]	Financial inclusion	14M wallets; 60+ countries	Mobile-first; stablecoin payments	Payments and savings only; no lending hierarchy or banking structure
R3 Corda [37]	Enterprise DLT	\$17B tokenized RWAs	Permissioned; consortium governance	Infrastructure layer only; no lending logic; closed access
World Bank Fund-sChain [39]	Dev. finance	250 projects by mid-2026	Hyperledger Besu; fund tracking	Tracks funds, does not implement lending or interest rate mechanics

Upon reviewing various projects a common feature was identified: a lending mechanism does not exist that operates with tiers, is distributed and facilitates transactions across different levels and within identical levels. Such innovation distinguishes the Crypto World Bank. Here's why it's different:

- Online lending in DeFi provides opportunities yet lacks a comprehensive framework. A defined structure is absent within the realm of DeFi lending.
- Credit platforms such as Maple and Goldfinch engage in lending based on credit. Credit-based lending is often focused on businesses and exists at just one tier. These platforms do not cater to personal loans.
- Payments are processed by banking systems such as Ripple. Lending and deposit services are not provided by this system. Ripple specializes in payment processing rather than other banking services.
- Celo provides assistance to individuals in emerging economies. Payment and savings functions are offered yet lending services remain absent.
- **Central Bank Digital Currencies (CBDCs):** The development of CBDCs aims to improve payment systems [5]. Concerns about privacy and control are raised as lending features are excluded. A lending layer does not exist on a CBDC. Users benefit from our platform which safeguards their interests while providing a level-based system similar to that utilized by CBDCs [45].

5.5 Risk Taxonomy

Table 5.5: Risk taxonomy and mitigation strategy.

Risk Category	Description	Severity	Mitigation
Partner non-cooperation	Key partners decline to participate	Medium	Initiate with low-barrier academic and NGO pilots
Smart contract vulnerability	Exploit in contract logic	High	OpenZeppelin primitives; formal audit (planned); pause mechanism
Regulatory adversity	Jurisdictional restrictions	Medium	Testnet-only prototype; regulatory sandbox engagement
AI/ML model degradation	Fraud detection accuracy decay	Low	Continuous retraining; human-in-the-loop; SHAP explainability

5.6 Technical Feasibility

Table 5.6: Technical feasibility assessment.

Component	Assessment	Evidence
Smart contracts	Fully feasible	Three contracts implemented, compiled, and tested with Hardhat (12+ passing unit tests); Solidity 0.8.20 with OpenZeppelin
Frontend DApp	Fully feasible	React 18 + TypeScript with all pages implemented; Wagmi and RainbowKit provide mature wallet integration
Blockchain deployment	Fully feasible	Polygon Amoy and Ethereum Sepolia provide zero-cost, production-equivalent environments
AI/ML integration	Feasible w/ constraints	Random Forest inference achieves sub-50ms latency; SHAP explanations computable in real-time
Database backend	Feasible	PostgreSQL schema designed (15 tables, 3NF); FastAPI provides async REST framework

5.7 Economic Feasibility

Table 5.7: Economic feasibility — zero-cost prototype.

Cost Category	Estimate	Notes
Blockchain deployment	\$0	Public testnets — no real cryptocurrency required
Frontend hosting	\$0	Vercel free tier or localhost for demo
Backend hosting	\$0	Render free tier or localhost
AI/ML training	\$0	Local machine (16 GB RAM, 16 GB VRAM) or Google Colab free tier
Development tools	\$0	Hardhat, VS Code, Git — all open-source
Total prototype cost	**\$0**	Entire prototype operates at zero financial cost

5.8 Revenue Projection

The following projection models annual revenue potential at full deployment scale. Calculations use a reference ETH price of **\$2,000** (February 2026 market rate) and interest rate parameters defined in Section 5.8.1.

Table 5.8: Revenue projection assumptions.

Parameter	Value
Reference ETH price	$2,000 (Feb 2026)
World Bank → National Bank rate	3% APR (wholesale inter-bank)
National Bank → Local Bank rate	5% APR (inter-bank)
Local Bank → Borrower rate	8% APR (retail lending)
Average loan term	12 months
Default rate provision	3% (conservative estimate)
Origination fee	0.25% per disbursement

Table 5.9: System-wide annual revenue summary.

Tier	Annual Revenue (ETH)	USD Equivalent
Tier 1: World Bank (1 entity)	31,525	$63,050,000
Tier 2: National Banks (5 entities)	25,775	$51,550,000
Tier 3: Local Banks (50 entities)	55,025	$110,050,000
Total platform revenue	**112,325**	**$224,650,000**
Borrower surplus generated	70,000–120,000	$140M–240M

Sources: *ETH price from spot market data (CoinGecko); interest rate tiers aligned with Aave [15] and Compound [16] DeFi benchmarks; default rate and origination fee from conservative industry estimates.*

Cost Model Considerations. Considerations for the cost model require careful thought. Important aspects need to be addressed by the team. Various factors will certainly influence the overall calculation:

- **Gas costs:** Gas expenses on Polygon PoS are quite low. A loan's entire process including request, approval, payment and 12 monthly installments results in costs under $0.15. Interest income amounts to less than 0.01% of our earnings. For Ethereum, loan costs may range from $50 to $750 influenced by network congestion. Processing transactions in batches or utilizing Layer 2 would be essential for the viability of small loans.
- **Infrastructure costs:** Backend hosting (API server, database, ML service), frontend delivery (CDN) and RPC nodes (Alchemy, Infura) can incur costs ranging between $500 and $2,000 monthly. An increase in expenses will be experienced with a rise in transactions.
- **Default losses:** A default rate of 3% is anticipated. Approximately $6.7 million annually is projected as the platform sees heavy usage. Anticipated earnings have already incorporated this amount.
- **Break-even analysis:** Opportunities for profit on loans as small as 0.01 ETH ($20) exist on Polygon PoS. Profitability on loans below 5 ETH ($10,000) is not achievable on Ethereum without Layer 2. Layer 2 is essential for facilitating small loans.

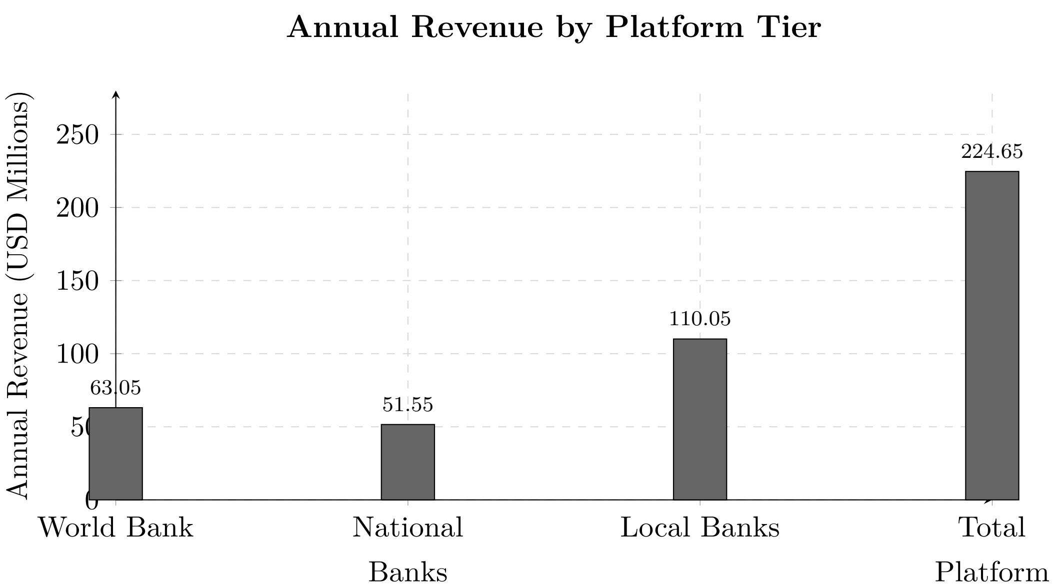

Figure 5.1: Annual revenue projection by tier (USD millions, at $2,000/ETH).

5.8.1 Transaction Economics: Interest Rates

Table 5.10: Interest rate parameters.

Parameter	Value	Benchmark
Base Annual Interest Rate	5–12% APR	Aligned with Aave/Compound variable rates
Rate Determination	Set by Local Bank approvers within World Bank-defined bounds	Configurable per-bank for local market conditions
Late Payment Penalty	2% of installment + 0.5%/week (capped at 10%)	Industry-standard late fee structure
Interest Calculation	Simple interest on outstanding principal	Transparent, borrower-friendly
Rate Transparency	All parameters stored on-chain	Publicly auditable; no hidden fees

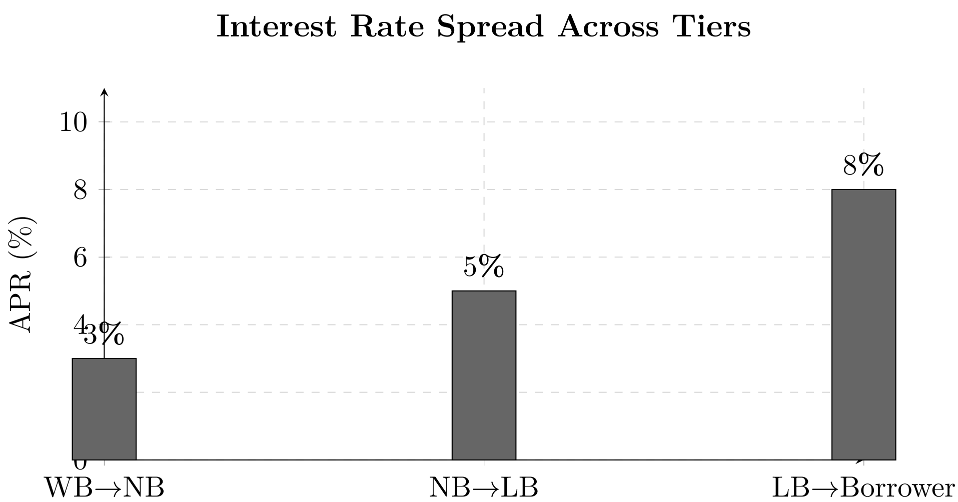

Figure 5.2: Hierarchical interest rate spread (APR) across the four-tier lending structure.

5.8.2 Global Economic Impact

Apart from revenue generated through platforms, various economic advantages are provided by the Crypto World Bank:

- **Capital deployment and fiscal multiplier:** Funding efforts have been made by the Crypto World Bank specifically targeting developing nations. A sum of $2 billion has been provided to nearly 1,000 borrowers. Loans from substantial institutions do not merely generate a single dollar; they generate approximately $2.5 to $3 in additional economic growth [19]. A significant increase in the economy estimated at $5 to $6 billion annually may be driven by lending practices. Approximately 16 new jobs are created in developing nations for every $1 million lent to small businesses as shown by IFC [20].
- **Remittance cost reduction:** Annually approximately $860 billion is transferred globally as remittances. A significant portion of this estimated between $48 and $56 billion is lost through transfer fees. Average fees are not less than 6.49% exceeding the United Nations target of 3% by more than double [26]. On-chain settlement, a rapid blockchain technique, is utilized by the Crypto World Bank to reduce costs to under 1%. Annually payments amounting to $55.6 billion are processed by Stellar incurring minimal fees around $0.0007 per transaction [44]. Payments are processed by Celo's MiniPay for under a cent across more than 60 nations [42].
- **Trapped capital liberation:** Trapped capital can now be released. Money is required by banks to remain in nostro and vostro accounts for international transactions [24]. Pre-funded accounts are no longer necessary due to the rapid payment settlements provided by the Crypto World Bank's blockchain system. Increased funds become available for lending purposes.
- **Financial inclusion:** Focused on financial inclusion, the platform caters to regular users. Features such as zero gas fees for transactions, optimized mobile performance, and the option to lend tiny sums below one dollar are provided. Credit remains uncomplicated and accessible for the 1.4 billion individuals globally who lack banking services [14]. A "leapfrog technology" is how decentralized finance (DeFi) has been described by the World Economic Forum serving to assist individuals in bypassing conventional banks [41]. In numerous developing nations mobile phones have overtaken landlines showcasing a significant shift in technology.
- **Transaction cost reduction:** Transaction expenses are significantly reduced. Fees on the Polygon PoS network remain below one cent resulting in over a **99% reduction** from the average $42 transaction fee seen in traditional correspondent banking systems [25]. Many small transactions which weren't feasible previously

are now possible due to this substantial decrease in costs.

- **Transparency as an economic good:** Clarity serves as an advantage. Reserve amounts, interest rates and transaction records are displayed on the blockchain. Harmful practices such as predatory lending are not enabled by this transparency; borrowers are less likely to face unfair charges or hidden fees. Information imbalance in traditional banking does not always favor borrowers particularly in developing countries resulting in disadvantageous agreements. Through the Crypto World Bank's platform, identical information can be viewed and verified by all participants promoting fairness and honesty in lending.

5.8.3 Value Proposition and Go-to-Market

Table 5.11: Go-to-market phases.

Phase	Activities	Timeline
Phase 1: Validation	Competition submission (BCOLBD 2025); thesis publication; open-source release	Current
Phase 2: Pilot	Regulatory sandbox application; institutional partnership; testnet-to-mainnet migration	6–12 months
Phase 3: Production	Multi-chain deployment; enhanced monitoring and analytics; governance token launch	12–24 months

5.9 Prototype Scope and Limitations

A straightforward version of the complete system is represented by the current prototype:

1. **Hierarchical lending scope:** Utilization of the Tier 1 World Bank Reserve contract is comprehensive. Deposits are managed, loan requests and approvals are processed alongside the display of reserve statistics. Not all functions of the Tier 2 (National Bank) and Tier 3 (Local Bank) contracts have been implemented. Role management and registration are not the only aspects focused on. Currently the process for fund transfers from the World Bank to National Bank and subsequently to Local Bank is incomplete. Four tiers will be included in the final design.

2. **Interest accrual and repayment:** Rules for interest rates (3%/5%/8%) are part of the system design. Fees for initiating loans and penalties for delays also exist. Interest calculations, payment schedules or penalties aren't managed automatically by current smart contracts. Future updates will incorporate these functions.
3. **InterBankLendingPool and multi-directional flows:** Plans for interbank loans at equal tier levels are included in the design (Section 1.7). Smart contracts have yet to be created for those components. Currently the prototype does not prioritize the upward transfer of funds.
4. **AI/ML integration:** AI tools such as fraud detection (Random Forest) and anomaly detection (Isolation Forest) along with decision explanations (SHAP) are set to be utilized by the system. A service has been established with FastAPI for machine learning. This service does not currently link to the loan approval process in the existing version.
5. **Backend architecture:** Express.js paired with MongoDB is utilized for rapid API development. The main database is intended to be PostgreSQL. FastAPI is utilized by the machine learning service.

Normal limits exist for the initial phase of the project. The design of the system and the establishment of basic infrastructure are prioritized during this stage. All features will not be included in this part. The complete system will be created in the subsequent phase.

Chapter 6

Conclusion

The Crypto World Bank deals with a **multi-party, multi-party coordination and trust** issue in hierarchical development finance through exploitation of the special features of blockchain technology: cryptographic, immutability, and programmable auditability. Our solution is better than the current DeFi lending protocols—which collectively manage over $55 billion of TVL [13] and everywhere use flat and single-tier pool architectures—through a four tier institutional hierarchy with cross-tier, same-tier and upward lending flows, along with a governance structure that deals with network membership, business operations and technology infrastructure.

The analysis of the competitive situation of 20+ existing projects in DeFi lending, institutional credit, banking infrastructure and financial inclusion affirm that no current system is a combination of multi-level lending (hierarchy), interbank lending mechanisms and graded access by borrowers in one decentralized architecture. The growing blockchain implementation in the institutions, as demonstrated by the World Bank FundsChain initiative [39], the multi-billion-dollar daily volumes of JPMorgan Kinexys [40] and R3 Corda's $17 billion in tokenized assets [37]—validates both the technical feasibility and institutional demand of blockchain-based financial infrastructure. The Crypto World Bank takes this trend of settlement and fund tracking to the fullest open and hierarchical lending structure.

The site is located on the border of institutional and decentralized finance lending, and financial inclusion—a mix, which is currently unaddressed in both the scholarly literature and the business blockchain environment. With a working prototype, a defined market and partnership plan, and a planned go-to-market plan, the Crypto World Bank is a possible and scalable addition to the new field of blockchain-based development finance.

The future work will be directed towards the following directions:

1. **Complete hierarchical lending implementation:** Complete the end-to-end wiring of cross-tier fund transfers between the World Bank Reserve, National Bank and Local Bank contracts, including automated capital distribution, interest accrual, generation of installments schedules, generation of cascading repayments enforcement as intended in the architecture.

2. **InterBankLendingPool and multi-directional flows:** Adopt the same-tier interbank lending pool and upward surplus repatriation mechanisms described in Section 1.7, so as to permit a full modeling of real-life banking liquidity flows in the decentralized structure.
3. **AI/ML pipeline integration:** Integrate the ML inference based on FastAPI service (Random Forest fraud detection, Isolation Forest anomaly detection, SHAP explainability) into the real-life loan approval process, integrating off-chain on-chain lending decisions to risk assessments.
4. **Mainnet and regulatory sandbox:** Moving off the testnet to the mainnet featuring real asset testing in regulated sandbox settings (e.g. UK FCA Digital Securities Sandbox, MAS FinTech Regulatory Sandbox Singapore, Bangladesh Bank FinTech Regulatory Sandbox).
5. **Extended surveillance assistance:** More developed risk-monitoring and anomaly detection modules can be considered in the following phases after the main lending workflow has been developed and blockchain architecture is completely stabilized.
6. **Cross-chain interoperability:** Ethereum Layer 2 multi-network lending rollups (Arbitrum, Optimism, Base) and other EVM chains to serve other geographic markets having optimized gas costs and local regulatory compliance.
7. **Formal security checking:** Formal verification of all smart contract code based on symbolic execution (Mythril), and static analysis (Slither), formal verification of property (Certora), especially multi-tier lending invariants such as cascading reserve ratio maintenance and cross-tier interest rate bound enforcement.
8. **Stablecoin integration, fiat on/off-ramps:** Support established blockchain coins (USDC, USDT, DAI) and native blockchain coins, with fiat on/off-ramping by collaborating with licensed money transmitters or anchor networks like MoneyGram that Stellar has implemented [44].
9. **Economic simulation and stress testing:** Agent-based simulation of the hierarchical lending system under market stress conditions (bank runs, cascading liquidations, correlated collateral devaluation) to prove convergence of interest rates, adequacy in the reserve ratios, and stability in the system.

Appendix A

Technology Stack

Table A.1: Technology stack summary.

Layer	Technology	Version / Notes
Smart Contract	Solidity, OpenZeppelin	0.8.20; Ownable, ReentrancyGuard
Frontend	React, TypeScript, Vite, Material-UI	React 18; MUI v5; Material Design 3
Wallet Integration	Wagmi, RainbowKit, Viem	EIP-1193 compliant
Build and Test	Hardhat	Automated test suite; deployment scripts
Backend API	Express.js, Node.js	REST API; middleware architecture
Database	PostgreSQL (designed)	15 tables; 3NF; relational integrity
Target Networks	Polygon Amoy, Ethereum Sepolia	Public testnets; zero-cost
AI/ML (planned)	Python, FastAPI, scikit-learn, SHAP	Random Forest, Isolation Forest, SHAP

Appendix B

Smart Contract Capabilities

The three smart contracts of the platform assist in the following operations:

- **World Bank Reserve Contract:** Reserving, deposit handling, national bank registration, fund allocation to the national banks, system pause/unpause, emergency withdrawal and world statistics.
- **National Bank Contract:** Local bank registration, borrowing the world bank reserve, fund allocation to local banks, and network stature queries.
- **Local Bank Contract:** Installment, loan approval and denial, loan request payments, management of bank users as well as approver designation.

References

[1] S. M. Werner, D. Perez, L. Gudgeon, A. Klages-Mundt, D. Harz, and W. J. Knottenbelt, “SoK: Decentralized Finance (DeFi),” *arXiv preprint arXiv:2101.08778*, 2022. [Online]. Available: https://arxiv.org/abs/2101.08778

[2] S. Bastankhah, M. Hashemi, and W. Shi, “An Adaptive Data-Driven DeFi Borrow-Lending Protocol,” *Proc. IEEE Int. Conf. Blockchain*, 2023.

[3] G. Palaiokrassas, A. Skoufis, and L. Tassiulas, “Leveraging Machine Learning for Multichain DeFi Fraud Detection,” *Proc. IEEE Int. Conf. Blockchain and Cryptocurrency (ICBC)*, 2023.

[4] D. Adom, E. O. Acheampong, and M. Boateng, “LIME and SHAP: A Comparison of Model-Agnostic Approaches to Explainability in Loan Approval Systems,” *International Journal of Advanced Computer Science and Applications*, vol. 13, no. 11, 2022.

[5] B. W. Tan, “Central Bank Digital Currency and Financial Inclusion,” *IMF Working Paper WP/23/69*, 2023. [Online]. Available: https://www.imf.org/en/Publications/WP

[6] F. T. Liu, K. M. Ting, and Z.-H. Zhou, “Isolation Forest,” in *Proc. IEEE Int. Conf. Data Mining (ICDM)*, pp. 413–422, 2008. DOI: https://doi.org/10.1109/ICDM.2008.17

[7] N. Atzei, M. Bartoletti, and T. Cimoli, “A Survey of Attacks on Ethereum Smart Contracts (SoK),” in *Proc. Int. Conf. Principles of Security and Trust (POST)*, pp. 164–186, 2017. DOI: https://doi.org/10.1007/978-3-662-54455-6_8

[8] S. M. Lundberg and S.-I. Lee, “A Unified Approach to Interpreting Model Predictions,” in *Advances in Neural Information Processing Systems (NeurIPS)*, pp. 4765–4774, 2017. [Online]. Available: https://proceedings.neurips.cc/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html

[9] P. Bracke, A. Datta, C. Jung, and S. Sen, “Machine Learning Explainability in Finance: An Application to Default Risk Analysis,” *Bank of England Staff Working Paper No. 816*, 2019. [Online]. Available: https://www.bankofengland.co.uk/working-paper/2019/machine-learning-explainability-in-finance

[10] R. Beck, C. Müller-Bloch, and J. L. King, “Governance in the Blockchain Economy: A Framework and Research Agenda,” *Journal of the Association for Information Systems*, vol. 19, no. 10, pp. 1020–1034, 2018. DOI: https://doi.org/10.17705/1jais.00518

[11] D. Mhlanga, “Blockchain Technology for Financial Inclusion and Sustainable Development,” in *Digital Financial Inclusion*, Springer, 2022.

[12] OpenZeppelin, "OpenZeppelin Contracts," 2024. [Online]. Available: https://openzeppelin.com/contracts

[13] DefiLlama, "Lending Protocols — DeFi TVL and Protocol Rankings," 2024–2025. [Online]. Available: https://defillama.com/protocols/lending

[14] World Bank, "The Global Findex Database 2021: Financial Inclusion, Digital Payments, and Resilience," 2021. [Online]. Available: https://www.worldbank.org/en/publication/globalfindex

[15] Aave, "Aave Protocol Documentation," 2024. [Online]. Available: https://docs.aave.com/

[16] Compound, "Compound Protocol Documentation," 2024. [Online]. Available: https://docs.compound.finance/

[17] BCOLBD 2025, "Blockchain Olympiad Bangladesh: Guideline and Evaluation Scheme," 2025. [Online]. Available: https://bcolbd.org/uploads/guideline/BLOCKCHAIN%20OLYMPIAD%20BANGLADESH%20Blockchain%20Guideline.pdf

[18] Galaxy Digital, "The State of Crypto Lending and Borrowing," Galaxy Research, 2024. [Online]. Available: https://www.galaxy.com/insights/research/the-state-of-crypto-lending

[19] World Bank, "World Development Report 2022: Finance for an Equitable Recovery," 2022. [Online]. Available: https://www.worldbank.org/en/publication/wdr2022

[20] International Finance Corporation (IFC), "MSME Finance Gap 2019," 2019. [Online]. Available: https://www.ifc.org/en/insights-reports/2019/msme-finance-gap

[21] Bank for International Settlements, "DeFi Lending: Intermediation Without Information?," BIS Working Paper No. 1183, 2024. [Online]. Available: https://www.bis.org/publ/work1183.pdf

[22] Deloitte, "Global Banking Outlook 2024," 2024.

[23] Michigan State University Libraries, "Literature Table and Synthesis — Nursing Literature Reviews," LibGuides, 2025. [Online]. Available: https://libguides.lib.msu.edu/nursinglitreview/table

[24] Committee on Payments and Market Infrastructures and Bank for International Settlements, "Correspondent Banking — Technical Report," CPMI Papers No. 147, 2016. [Online]. Available: https://www.bis.org/cpmi/publ/d147.pdf

[25] Ripple, "RLUSD Use Case Analysis," XRP Academy, 2026. [Online]. Available: https://xrpacademy.com/blog/rlusd-use-case-analysis-calendar-570

[26] World Bank, "Remittance Prices Worldwide: Making Markets More Transparent," Migration and Development Brief 40, 2024. [Online]. Available: https://remittanceprices.worldbank.org/

[27] DefiLlama, "Aave V3 TVL, Fees, Revenue & Income Statement," March 2026. [Online]. Available: https://defillama.com/protocol/aave-v3

[28] World Inequality Lab, "Exorbitant Privilege," *World Inequality Report 2026*, 2026. [Online]. Available: https://wir2026.wid.world/insight/exorbitant-privilege/

[29] DefiLlama, "Compound V3 TVL, Fees, Revenue & Income Statement," March 2026. [Online]. Available: https://defillama.com/protocol/compound-v3

[30] Fensory, "Sky Protocol Projects $611M Revenue in 2026 as USDS Supply Targets $20.6 Billion," February 2026. [Online]. Available: https://www.fensory.com/intelligence/rwa/sky-protocol-tokenization-regatta-solana-february-2026

[31] Fensory, "Maple Finance — Institutional DeFi Lending Protocol," 2026. [Online]. Available: https://www.fensory.com/insights/protocols/maple-finance

[32] Fensory, "DeFi Credit Platforms Hit $2.4B Milestone," March 2026. [Online]. Available: https://www.fensory.com/intelligence/rwa/defi-private-credit-platforms-2-4-billion-lo

[33] Financial Stability Board, "Enhancing Cross-border Payments: Stage 3 Roadmap," 2020. [Online]. Available: https://www.fsb.org/2020/10/enhancing-cross-border-payments-stage-3-r

[34] A. Beyer, B. Gasperini, and P. Theodoridis, "Monetary Policy and Inequality: Distributional Effects of Asset Purchase Programs," *Journal of International Money and Finance*, vol. 157, 2025.

[35] Federal Reserve Board, "Monetary Policy and the Distribution of Income: Evidence from U.S. Metropolitan Areas," FEDS Notes, March 2025. [Online]. Available: https://www.federalreserve.gov/econres/notes/feds-notes/monetary-policy-and-the-distribution-of-i html

[36] Mises Institute, "The Theory of the Bottom 99 Percent: The Cantillon Effect," 2026. [Online]. Available: https://mises.org/mises-wire/theory-bottom-99-percent-cantillon-effect

[37] GlobeNewswire, "R3's Corda Leads Tokenized RWA Market with Over $10 Billion in On-chain Assets," February 2025. [Online]. Available: https://www.globenewswire.com/news-release/2025/02/13/3025637/

[38] Centrifuge, "Real-World Asset Tokenization: Key Trends from 2025," 2025. [Online]. Available: https://centrifuge.io/blog/real-world-asset-tokenization-trends-2025

[39] World Bank, "World Bank Group Tracks Project Funds with New Blockchain Tool," Press Release, September 2025. [Online]. Available: https://www.worldbank.org/en/news/press-release/2025/09/26/world-bank-group-tracks-project-funds-with-new-blockcha

[40] JPMorgan, "Kinexys Digital Payments: Real-Time Multicurrency Payments," 2025. [Online]. Available: https://www.jpmorgan.com/onyx/coin-system.htm

[41] World Economic Forum, "Why Decentralized Finance Is a Leapfrog Technology for the 1.1 Billion People Who Are Unbanked," September 2022. [Online]. Available: https://weforum.org/stories/2022/09/decentralized-finance-a-leapfrog-technology-for-the-unbar

[42] Opera Newsroom, "160M CELO Allocation Proposal to Grow Opera from Distribution Partner into Key Network Stakeholder," March 2026. [Online]. Available: https://press.opera.com/2026/03/19/opera-celo-partnership-2026/

[43] Morpho, "Network Data," March 2026. [Online]. Available: https://data.morpho.org/

[44] Stellar, "End of Year 2025 Report — Execution at Scale," 2025. [Online]. Available: https://www.stellar.org/blog/foundation-news/2025-year-in-review

[45] Human Rights Foundation, "Tracking CBDCs Before They Track You," 2025. [Online]. Available: https://hrf.org/latest/tracking-cbdcs-before-they-track-you

[46] Y. Li, X. Zhang, and H. Wang, "Design and Implementation of a Multi-Chain Lending Model in Blockchain," in *Proc. IEEE Int. Conf. Blockchain*, 2024. DOI: https://doi.org/10.1109/Blockchain62396.2024.10729983

[47] R. Xu, S. Chen, and L. Yang, "An Evaluation System for DeFi Lending Protocols," in *Proc. IEEE Int. Conf. Blockchain and Cryptocurrency (ICBC)*, pp. 1–6, 2023. DOI: https://doi.org/10.1109/ICBC56567.2023.10240601

[48] A. Sharma, R. K. Singh, and S. Gupta, "Blockchain Empowered Framework for Peer to Peer Lending," in *Proc. IEEE Int. Conf. Blockchain Computing and Applications (BCCA)*, pp. 142–147, 2021. DOI: https://doi.org/10.1109/BCCA53669.2021.9596552

[49] M. T. Hassan, F. Ahmad, and A. Mehmood, "Blockchain and Machine Learning for Fraud Detection: A Privacy-Preserving and Adaptive Incentive Based Approach," *IEEE Access*, vol. 10, pp. 87115–87131, 2022. DOI: https://doi.org/10.1109/ACCESS.2022.3199498

[50] Y. Wang, J. Liu, and Z. Chen, "ContractWard: Automated Vulnerability Detection Models for Ethereum Smart Contracts," *IEEE Trans. Network Science and Engineering*, vol. 8, no. 2, pp. 1133–1144, 2020. DOI: https://doi.org/10.1109/TNSE.2020.2968505

[51] C.-F. Liao, T.-H. Tsai, and C.-J. Chen, "SoliAudit: Smart Contract Vulnerability Assessment Based on Machine Learning and Fuzz Testing," in *Proc. IEEE Int. Conf. Internet of Things (iThings)*, pp. 458–465, 2019. DOI: https://doi.org/10.1109/iThings/GreenCom/CPSCom/SmartData.2019.00098

[52] S. So, M. Lee, J. Park, H. Lee, and H. Oh, "VERISMART: A Highly Precise Safety Verifier for Ethereum Smart Contracts," in *Proc. IEEE Symp. Security and Privacy (S&P)*, pp. 1678–1694, 2020. DOI: https://doi.org/10.1109/SP40000.2020.00032

[53] S. Islam, R. Khan, and M. A. Rahman, "Central Bank Digital Currency (CBDC): Design Requirements and Challenges," in *Proc. IEEE Int. Conf. Computing, Communication, and Intelligent Systems (ICCCIS)*, 2024. DOI: https://doi.org/10.1109/ICCCIS62002.2024.10634472

[54] M. S. Alam, S. M. Hossain, and A. K. Das, "Towards Using Blockchain Technology for Microcredit Industry in Bangladesh," in *Proc. IEEE Region 10 Symp. (TENSYMP)*, pp. 1–6, 2021. DOI: https://doi.org/10.1109/TENSYMP52854.2021.9392730

[55] P. Tolmach, Y. Li, and S. Lin, "Optimal Gas Fee Minimization in DeFi: Enhancing Efficiency and Security on the Ethereum Blockchain," *IEEE Trans. Dependable and Secure Computing*, 2024. DOI: https://doi.org/10.1109/TDSC.2024.3495637

[56] S. Gupta, A. Kumar, and R. Sharma, "SHAP-based Interpretable Models for Credit Default Assessment Using Machine Learning," in *Proc. IEEE Int. Conf. Artificial Intelligence and Data Science*, 2024. DOI: https://doi.org/10.1109/ICAIDS60875.2024.10840375